



E2 - Getting Started with the Development Ecosystem for PSOC™ Edge E84



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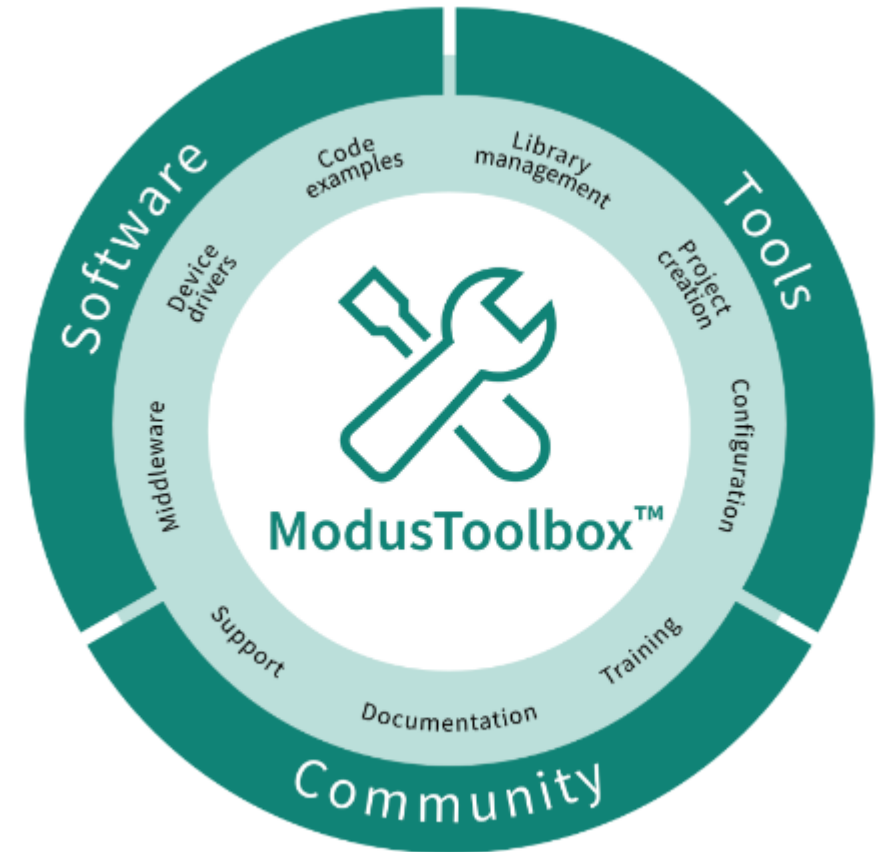
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PSOC™ Edge E84 Resources

- PSOC™ Edge E84 MCU Product page and PSOC™ Developer Journey at Infineon.com
- Datasheets
- Reference manuals (RM): architecture and registers
- Application notes
- Code examples
- Development kits
- KBAs and community forum
- PSOC™ Edge E84 MCU Product page and PSOC™ Developer Journey at Infineon.com

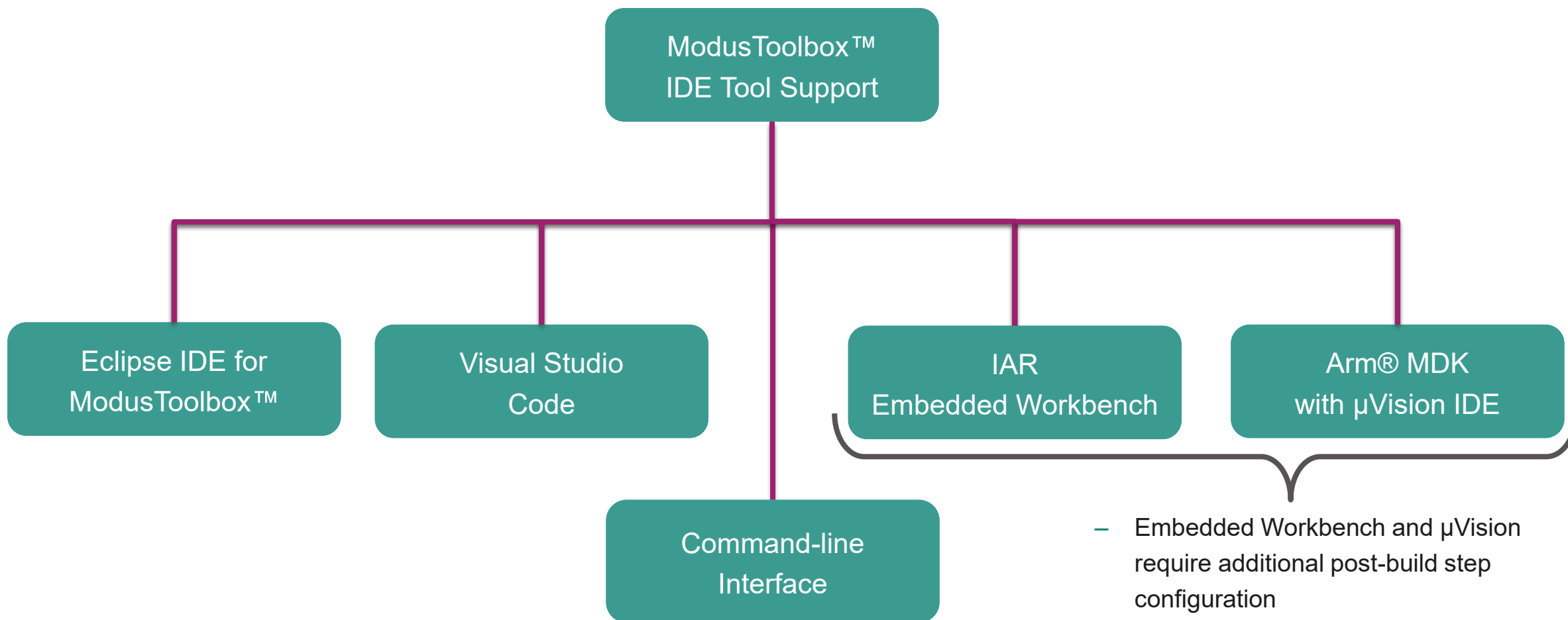
Firmware / Application Development

- **ModusToolbox™ provides**
 - Development environment for Infineon MCUs
 - Flexible set of tools for project and system level configuration
 - Low-level drivers
 - Libraries
 - Operating system support (Linux, macOS, Windows)



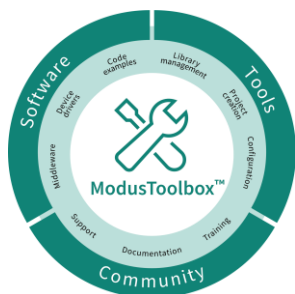
Reference link : <https://www.infineon.com/ModusToolboxInstallguide>

Support for IDEs



- Embedded Workbench and μVision require additional post-build step configuration
- Details documented within IDE respective user manuals

Understanding ModusToolbox™ and Zephyr



- Provides a **targeted ecosystem** focused on development of **Infineon devices**.

- IDE support, including **Eclipse IDE**, **VS Code**, and **3rd party IDEs**.
- Support for GCC, IAR, and Arm compilers using **'make' based ModusToolbox™ build system**.
- Graphical **UI configurations tool**
- Leverages **Infineon developed** low-level drivers, middleware abstraction, middleware, and code examples.
- Enables key partner middleware via **ModusToolbox™ & Friends** program

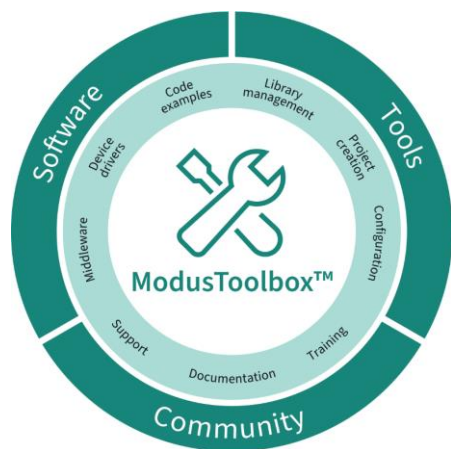


Zephyr®

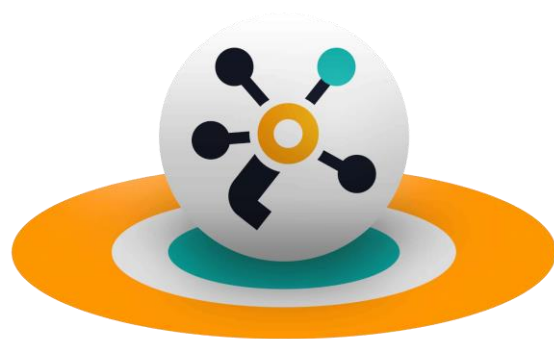
- Provides a **community driven ecosystem** with support for a targeted set of MCUs from a **wide range of silicon providers**.

- Targeted at **command-line users** with some community support for **VS Code**.
- Default support for GCC compiler using **'west' based Zephyr build environment**. Additional toolchains are now being supported.
- Text-based configuration via **devicetree and Kconfig**.
- Leverages vendor provided device support; Infineon support based on **low-level drivers**.
- **Community contributed** middleware, with support for Zephyr defined driver API, stacks, and RTOS.

Overview of DEEPCRAFT™ Studio and ModusToolbox™ workflow



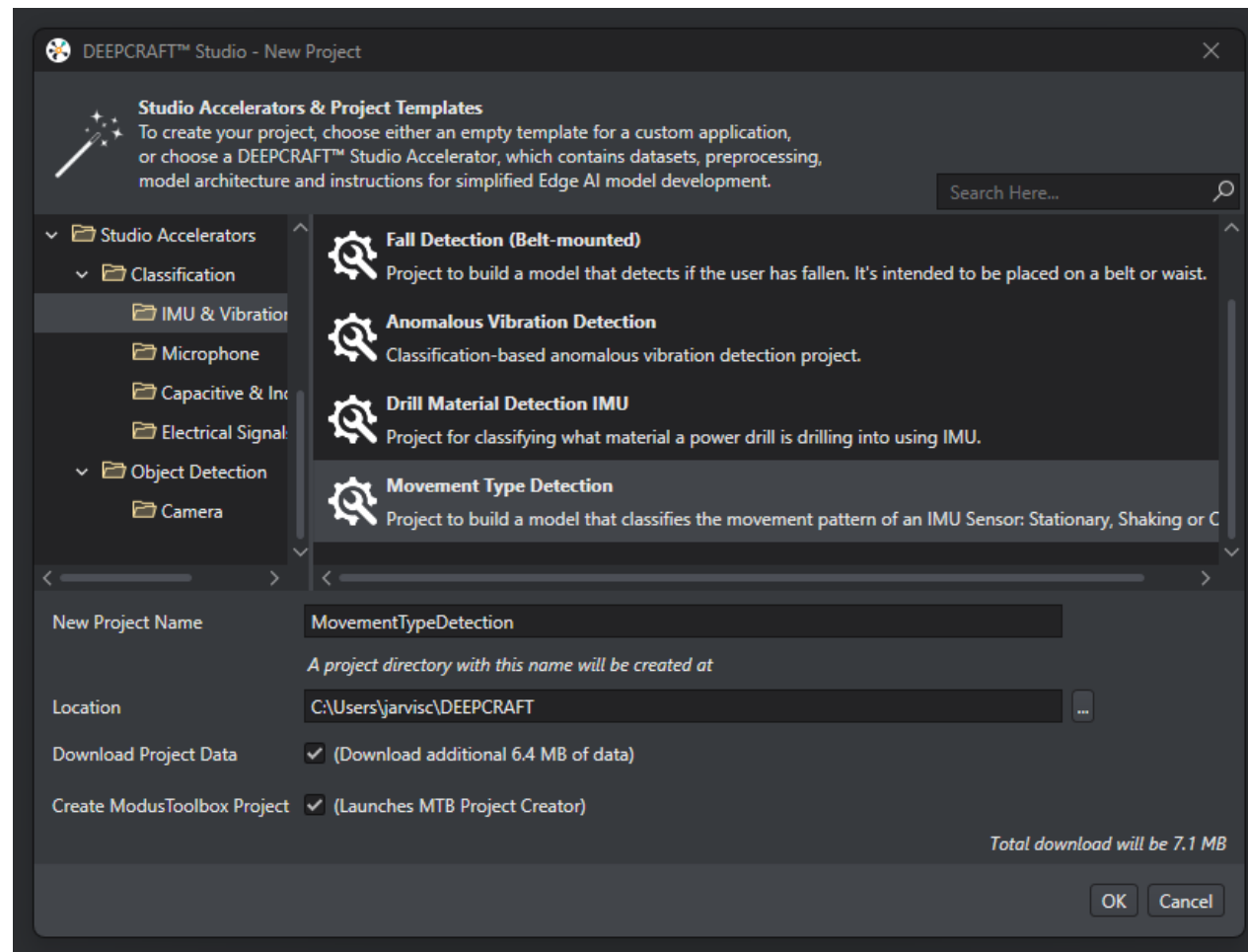
- Installation of ModusToolbox™ and DEEPCRAFT™ supported within the **ModusToolbox™ Setup** program
- Code Examples aligned with DEEPCRAFT™ Studio **Accelerator projects** for IMU, Vibration, and Microphone
- Code Examples aligned with DEEPCRAFT™ Studio **Data Collection with Graph UX**
- Code Examples aligned with DEEPCRAFT™ Model Converter **Profiling**
- **Project connector** files in ModusToolbox™ applications with Quick Panel shortcuts for linking to DEEPCRAFT™ Studio Projects



- Integration of ModusToolbox™ Project Creator, supporting coordinated DEEPCRAFT™ Studio and ModusToolbox™ project creation
- Integration ModusToolbox™ Machine Learning Core Tools (ml-coretools), leveraged during Infineon device specific code generations and optimization
- ModusToolbox™ project detection and selection during model code generation

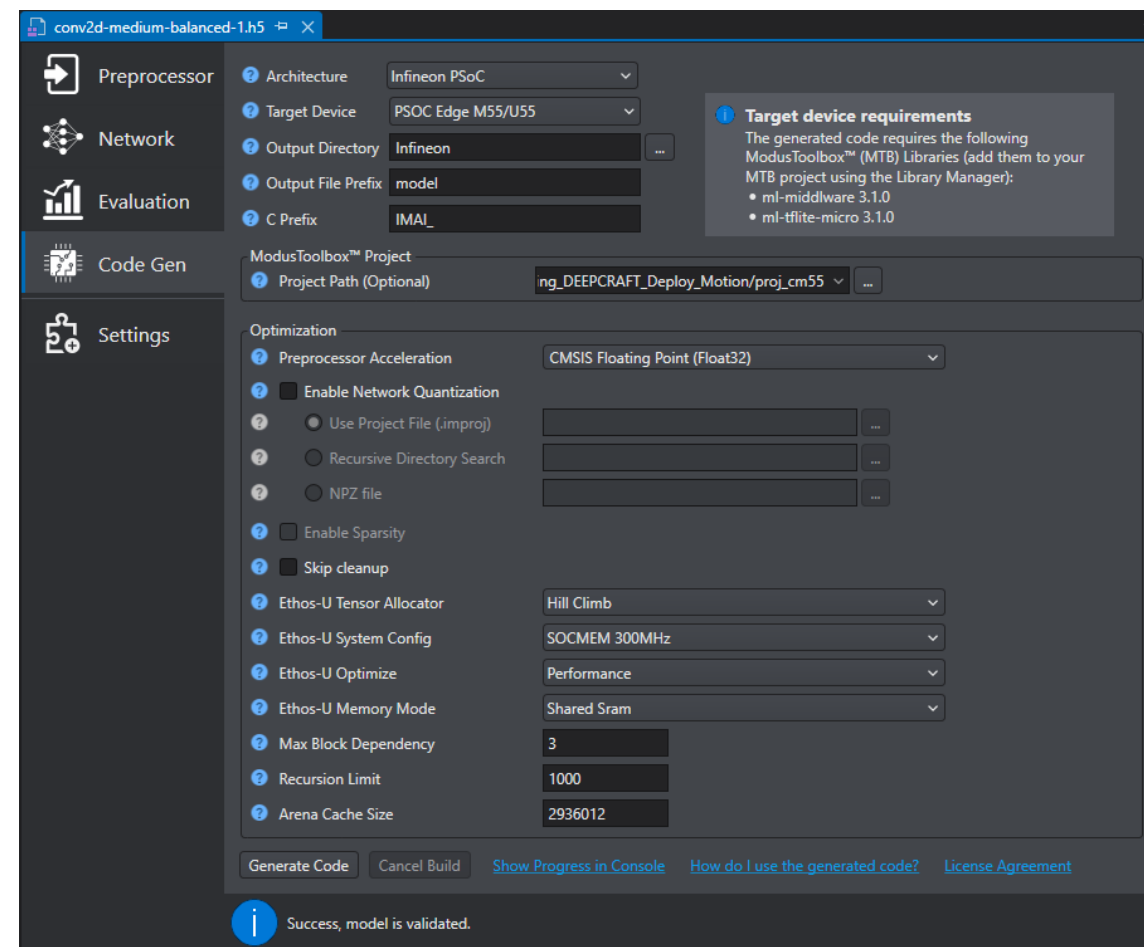
Code Examples with DEEPCRAFT™ and ModusToolbox™

- When creating a new DEEPCRAFT™ Studio project, a ModusToolbox™ project can be created at the same time.
- DEEPCRAFT™ Studio includes Studio Accelerator projects that align with ModusToolbox™ code examples, enabling models generated from Studio to be more easily leveraged within a ModusToolbox™ application.
- ModusToolbox™ code examples include DEEPCRAFT™ Deploy Audio and Deploy Motions applications.



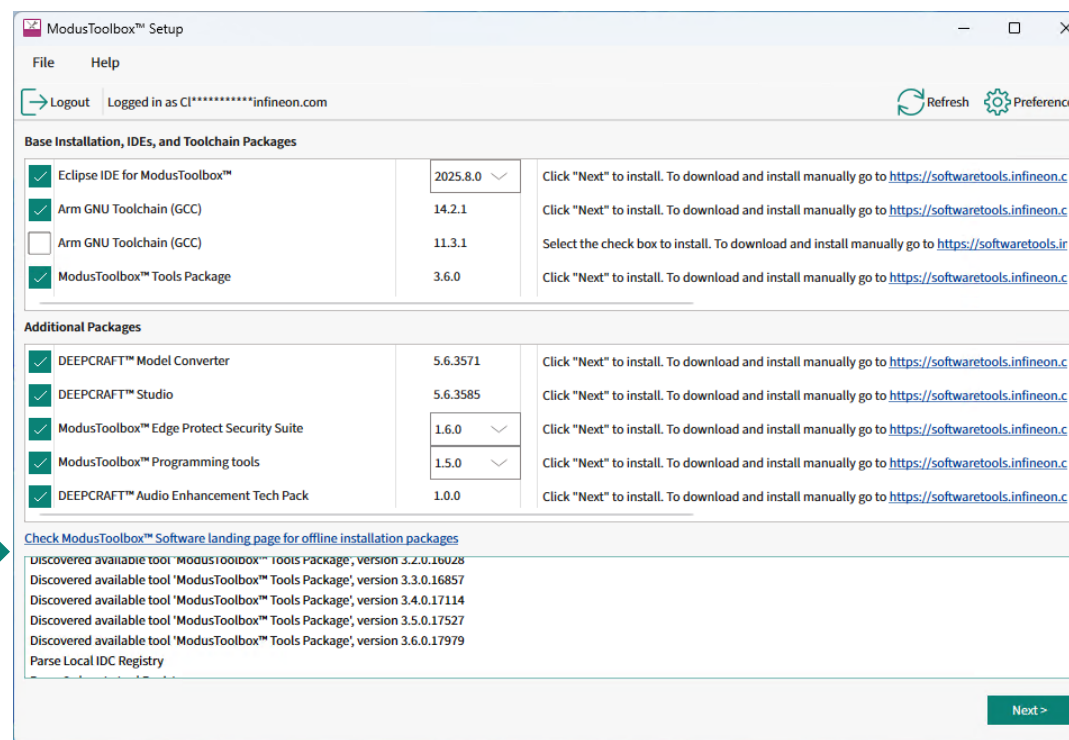
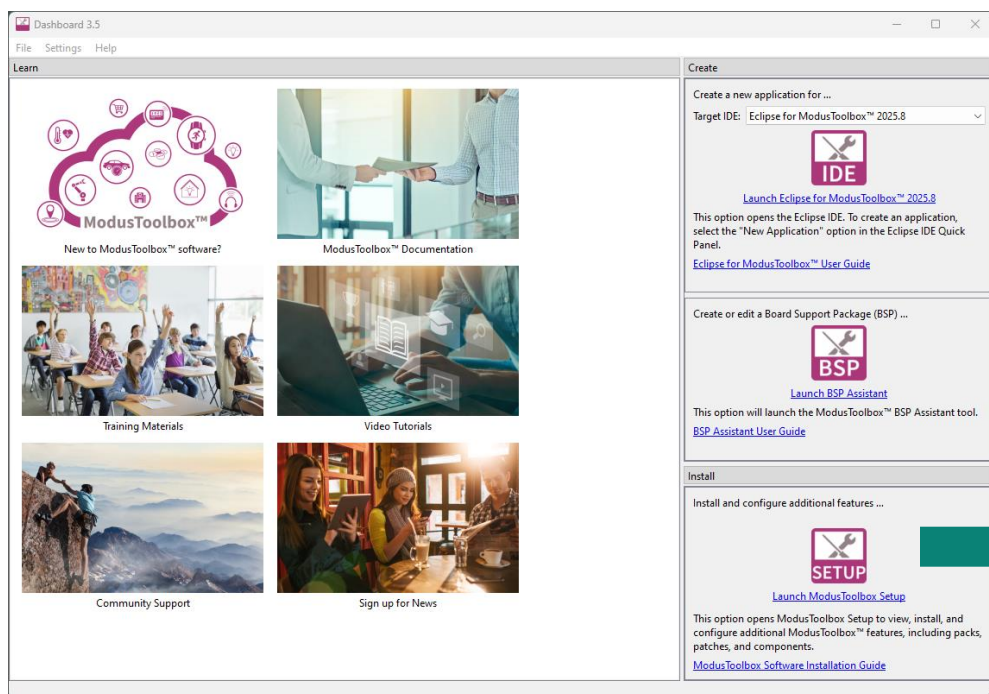
DEEPCRAFT™ Studio Model Code Generation

- DEEPCRAFT™ Studio code generation dialog includes pre-defined architecture selections for Infineon devices, including PSOC™ Edge E84
- DEEPCRAFT™ Studio code generation integrates the ModusToolbox™ Machine Learning Core Tools (ml-coretool) with integrated Vela compiler supporting PSOC™ Edge E84.
- Selection of Infineon device enables project selection of the connected ModusToolbox™ project and core specific project path.
- Selected ModusToolbox™ project path will be used during the code generation step to enable iterative model development and updates.



Installation support within ModusToolbox™ Setup

- ModusToolbox™ Setup program can be used to install and manage DEEPCRAFT™ applications, including Studio, Model Converter, and Audio Enhancement Tech Pack



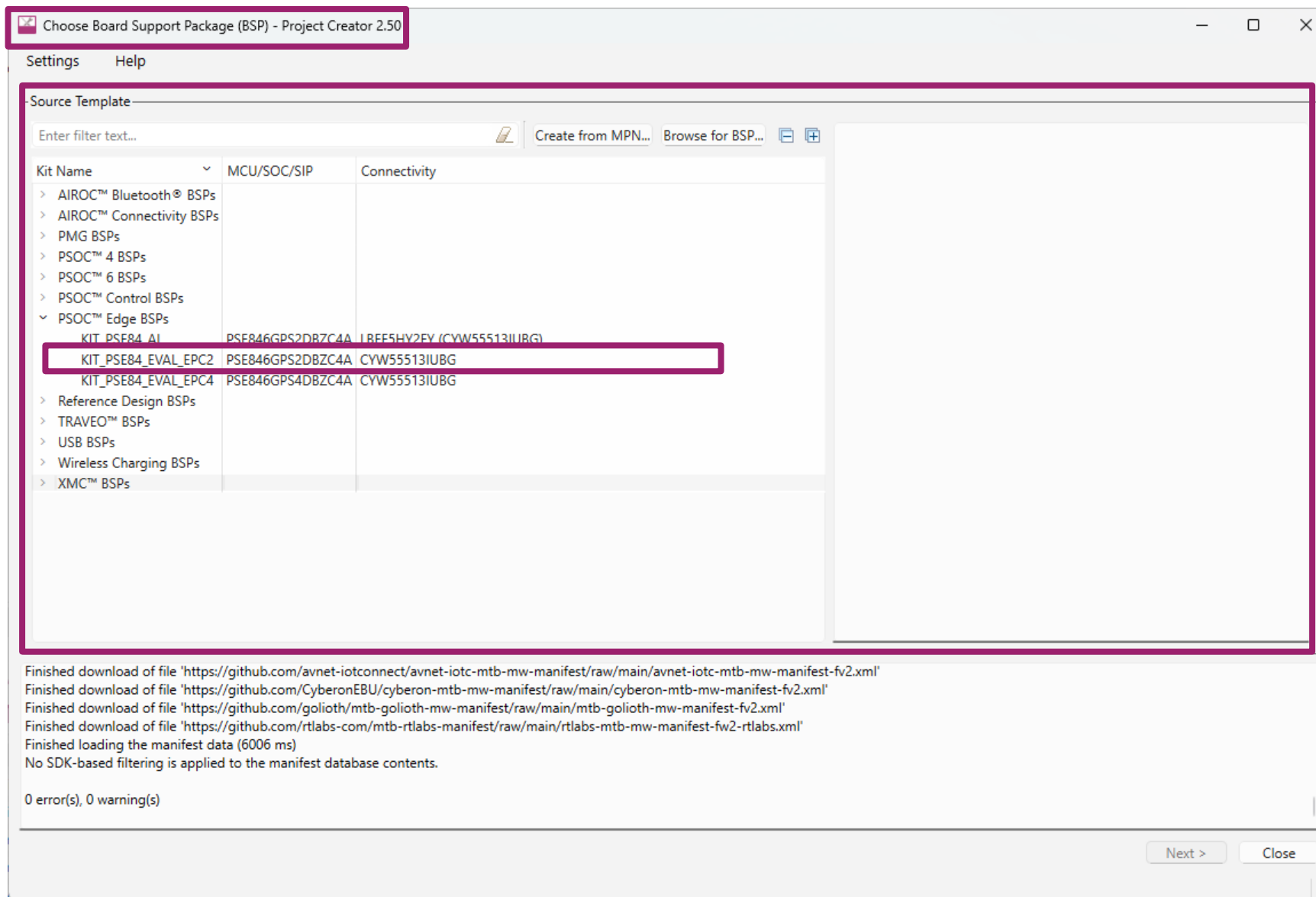
Tools utilized in this presentation

- ModusToolbox™ v3.6
- Eclipse IDE for ModusToolbox™ v2025.4.0
- Edge Protect Security Suite v1.6
- ModusToolbox™ Programming Tools v1.5.0
- KIT_PSE84_EVAL_EPC2 BSP v1.0.0
- Terminal Emulator
- KIT_PSE84_EVAL revG

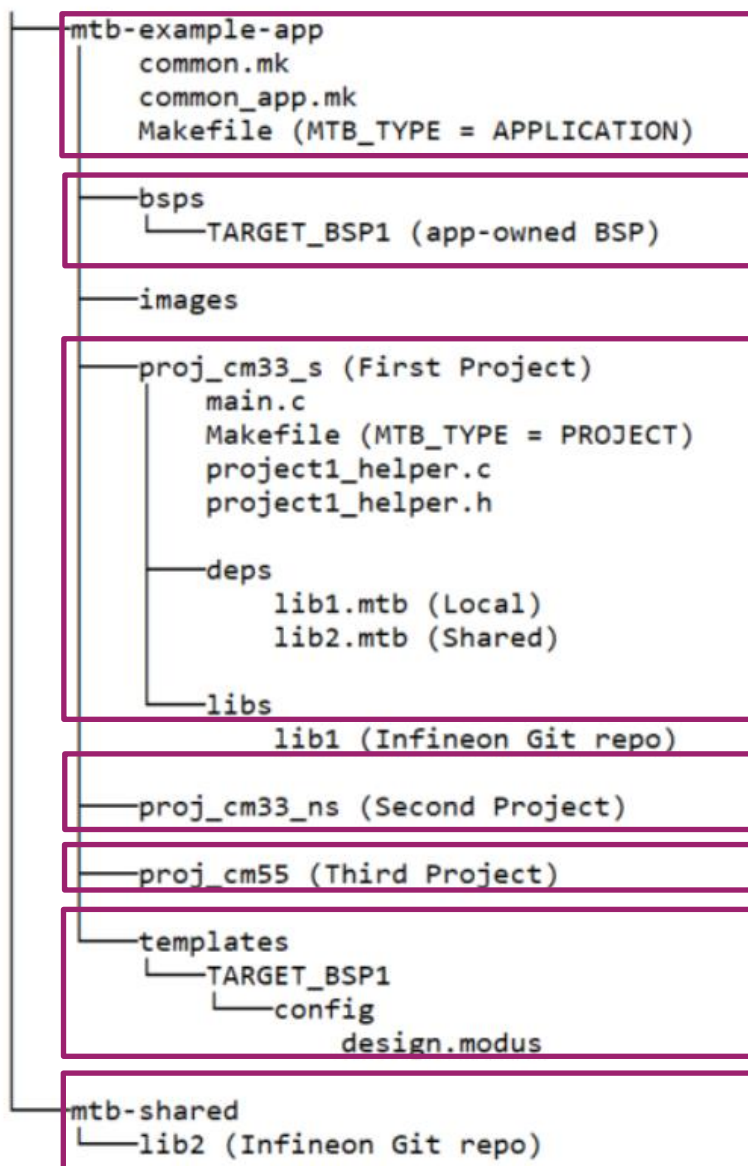
Key documentation leveraged in this video

- **AN235935** – Getting started with PSOC™ Edge E84 on ModusToolbox™ software
- ModusToolbox™ PSE8xxGP Device Support Library (DSL)

Project Creator



Understanding the Application Structure



Library Manager

Library Manager 2.50
Settings
Help

Application Directory: C:/Workspace/PSOC_Edge_Hello_World
Browse...

Enter filter text...

Name	Update Available	Remove
BSPs APP_KIT_PSE84_EVAL_EPC2 (ACTIVE)		
proj_cm33_ns Libraries async-transfer bt-fw-ifs-cyw55500a1 cmsis core-lib core-make device-db mtb-dsl-pse8xxgp mtb-ipc mtb-srf retarget-io se-rt-services-utils		
proj_cm33_s Libraries async-transfer bt-fw-ifs-cyw55500a1 cmsis core-lib		

Add BSP
Add Library

Properties
README.md
RELEASE.md

Property	Value
Name	mtb-dsl-pse8xxgp
Version	1.1.1 release
Type	Shared Git Repo
Location	C:/Workspace/mtb_shared/mtb-dsl-pse8xxgp/release-v1.1.1

mtb-dsl-pse8xxgp

Device support library for Explorer PSE8XXGP devices.

Version details: 1.1.1 release

Reference Documentation:

(stdlib-stubs)

PSOC E8XXGP Device Support Library

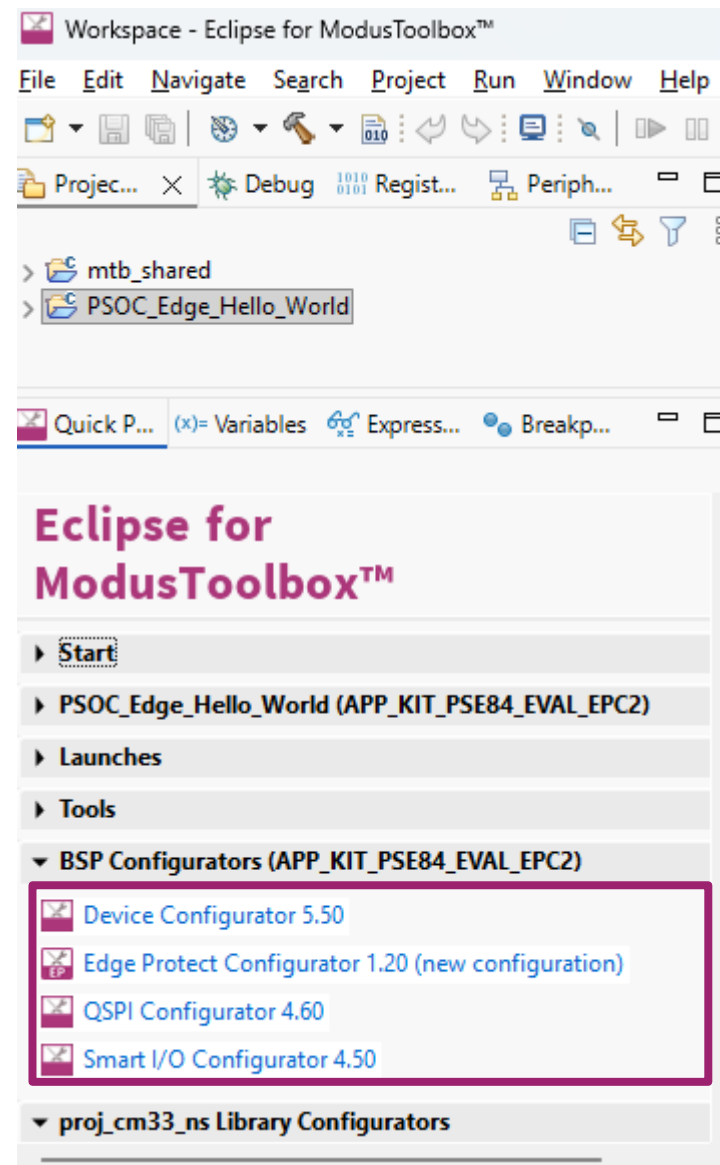
Syspm-Callbacks

Starting to refresh dependencies for project 'proj_cm33_ns'.
Refreshed all dependencies for project 'proj_cm33_ns'.
Starting to refresh dependencies for project 'proj_cm33_s'.
Refreshed all dependencies for project 'proj_cm33_s'.
Starting to refresh dependencies for project 'proj_cm55'.
Refreshed all dependencies for project 'proj_cm55'.
0 error(s), 0 warning(s)
Successfully acquired the information.

Update
Close

BSP Configurators

- Device Configurator
 - QSPI Configurator
 - Smart I/O Configurator
 - Edge Protect Configurator
-
- Configurators
 - Creates structures needed for peripherals used by PDL



Device Configurator

C:/Workspace/PSOC_Edge_Hello_World/bsps/TARGET_APP_KIT_PSE84_EVAL_EPC2/config/design.modus - Device Configurator 5.50

File Edit Settings View Help

PSE846GPS2DBZC4A CYW55513IUBG

Peripherals Analog Pins Clocks System Memory DMA

Enter filter text...

Resource	Name(s)	Personality
Communication		
Controller Area Network FD (CAN FD) 0		
☐ Ethernet 0	eth_0	
☒ I3C	:YBSP_I3C_CONTROLLER	I3C Controller-1.0
Quad Serial Memory Interface (QSPI) 0		
Quad Serial Memory Interface (QSPI) 1		
☒ SD Host Controller (SDHC) 0	CYBSP_WIFL_SDIO	SD Host Controller-1.0
☒ SD Host Controller (SDHC) 1	CYBSP_SDHC_1	SD Host Controller-1.0
☒ Serial Communication Block (SCB) 0	:YBSP_I2C_CONTROLLER	I2C-4.0
☐ Serial Communication Block (SCB) 1	scb_1	
☒ Serial Communication Block (SCB) 2	CYBSP_DEBUG_UART	UART-3.0
☐ Serial Communication Block (SCB) 3	scb_3	
☒ Serial Communication Block (SCB) 4	CYBSP_BT_UART	UART-3.0
☒ Serial Communication Block (SCB) 5	CYBSP_EZ_I2C_TARGET	EZ_I2C-3.0
☐ Serial Communication Block (SCB) 6	scb_6	
☐ Serial Communication Block (SCB) 7	scb_7	
☐ Serial Communication Block (SCB) 8	scb_8	
☐ Serial Communication Block (SCB) 9	scb_9	
☒ Serial Communication Block (SCB) 10	:YBSP_SPI_CONTROLLER	SPI-3.0
☐ Serial Communication Block (SCB) 11	scb_11	
Time Division Multiplexing (TDM)		
☒ Universal Serial Bus (USB) 0	CYBSP_USB_DEVICE_0	emUSB-Host/Device-1.0
Digital		
System		

Serial Communication Block (SCB) 2 (CYBSP_DEBUG_UART) - Parameters

Enter filter text...

Name	Value
Overview	
Configuration Help	Open UART (SCB) Documentation
General	
Com Mode	Standard
Baud Rate (bps)	115200
Oversample	10
Bit Order	LSB First
Data Width	8 bits
Parity	None
Stop Bits	1 bit
Enable Digital Filter	☐
Support RS-485	
TX-Enable	☐
Flow Control	
Enable Flow Control	☐
CTS Polarity	Active Low
RTS Polarity	Active Low
RTS Activation Level	63
Connections	
Clock	☒ 16 bit Divider 1 clk (CYBSP_DEBUG_UART_CLK_DIV) [USED]
RX	☒ P6[5] digital_inout (CYBSP_DEBUG_UART_RX) [USED]
TX	☒ P6[7] digital_inout (CYBSP_DEBUG_UART_TX) [USED]
RX Trigger Output	<unassigned>

Code Review

Enter search text...

```

/*
 * NOTE: This is a preview only. It combines elements of the
 * cycfg_peripherals.c and cycfg_peripherals.h files located in the folder:
 * C:/Workspace/PSOC_Edge_Hello_World/bsps/TARGET_APP_KIT_PSE84_EVAL_EPC2/config/GeneratedSource
 */

```

Device Configurator contd...

FileEditSettingsViewHelp

PSE846GPS2DBZC4ACYW55513IUBG

PeripheralsAnalogPinsClocksSystemMemoryDMA

Enter filter text...

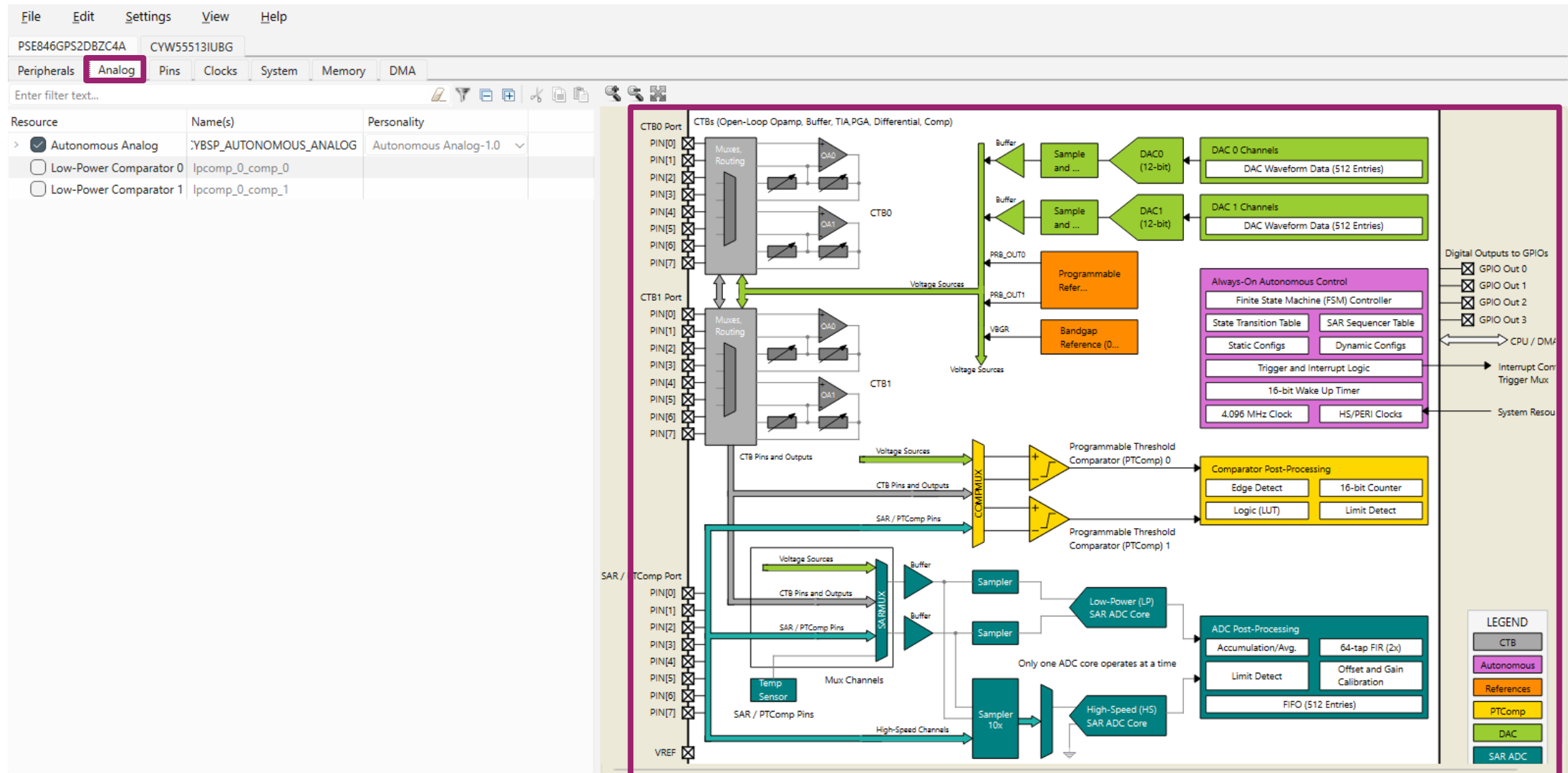
Resource	Name(s)	Personality
> Port 0		
> Port 1		
> Port 2		
> Port 3		
> Port 4		
> Port 5		
> Port 6		
> Port 7		
> Port 8		
> Port 9		
> Port 10		
> Port 11		
> Port 12		
> Port 13		
> Port 14		
> Port 15		
▼ Port 16		
☒ P16[0]	CYBSP_SPI_CLK	Pin-3.0 ▼
☒ P16[1]	CYBSP_SPI_MOSI	Pin-3.0 ▼
☒ P16[2]	CYBSP_SPI_MISO	Pin-3.0 ▼
☒ P16[3]	CYBSP_SPI_CS	Pin-3.0 ▼
☐ P16[4]	CYBSP_USB_FAULT	
☒ P16[5]	CYBSP_USER_LED3, CYBSP_LED_BLUE	Pin-3.0 ▼
☒ P16[6]	CYBSP_USER_LED2, CYBSP_LED_GREEN	Pin-3.0 ▼
☒ P16[7]	CYBSP_USER_LED1, CYBSP_USER_LED, CYBSP_LED_RED	Pin-3.0 ▼
> Port 17		
> Port 18		
> Port 19		
> Port 20		
> Port 21		

151413121110987654321

PSE846GPS2DBZC4A (BGA-220)

☒ Assigned☐ Power☒ Error
☒ Dedicated☐ GPIO

Device Configurator contd...



Device Configurator contd...

File Edit Settings View Help

PSE846GPS2DBZC4A CYW55513IUBG

Peripherals

Analog

Pins

Clocks

System

Memory

DMA

Enter filter text...

Resource	Name(s)	Personality
☒ Protection Domains ☒ Memory Protection Controller (MPC) Memory Protection Units (MPU) Peripheral Protection Controllers (PPC) Security Attribution Units (SAU)	vres_0_protection_0	Protection-1.0
	vres_0_protection_0_mp...	MPC-1.0
☒ System Debug Power ☒ Power	debug600_0	Debug Access-2.0
	srss_0_power_0	Power Settings-1.1

Power - Parameters

Enter filter text...

Name	Value
Overview	
? SysPm API Reference Open SysPm Documentation	
General	
? System Active Power Profile Power Profile - HP (High Performance Profile - vccd = 0.9V, vccsram = 0.9V)	
Power domains	
? PD1 power domain ☒	
System Idle Parameters	
? System Idle Power Mode System Deep Sleep	
? Deep Sleep Latency (ms) 20	
Operating Conditions	
? Core Buck Voltage and Mode 0.9V High Power	
? VDDA Voltage (mV) 1800	
? VDDD Voltage (mV) 1800	
? VDDIO0 Voltage (mV) 1800	
? VDDIO1 Voltage (mV) 1800	
Wakeup Pins	
Code Preview	
Enter search text...	
<pre> /* * NOTE: This is a preview only. It combines elements of the * cycfg_system.c and cycfg_system.h files located in the folder: * C:/Workspace/PSOC_Edge_Hello_World/bapa/TARGET_APP_KIT_PSE84_EVAL_EPC2/config/GeneratedSource </pre>	

Device Configurator contd...

File Edit Settings View Help
PSE846GPS2DBZC4A CYW55513IUBG
Peripherals Analog Pins **Clocks** System Memory DMA
Enter filter text...

Resource	Name(s)	Personality
> Clocks		
> Dividers		
> High Frequency		
> Peripheral		
> Peri 0		
> Peri 0 Clock Group 1		
> 8 bit		
> 16 bit		
16 bit Divider 0	CYBSP_I2C_CONTROLLER_CLK_DIV	Peripheral Clock-1.1
16 bit Divider 1	CYBSP_DEBUG_UART_CLK_DIV	Peripheral Clock-1.1
16 bit Divider 2	YBSP_GENERAL_PURPOSE_TIMER_CLK_DIV	Peripheral Clock-1.1
16 bit Divider 3	CYBSP_PWM_LED_CTRL_CLK_DIV	Peripheral Clock-1.1
> 16.5 bit		
> 24.5 bit		
> Peri 0 Clock Group 2		
> 8 bit		
> Peri 0 Clock Group 3		
> 8 bit		
> Peri 0 Clock Group 4		
> 8 bit		
> Peri 0 Clock Group 7		
> 16.5 bit		
> Peri 0 Clock Group 8		
> 16 bit		
16 bit Divider 0	peri_0_group_8_div_16_0	
> Peri 0 Clock Group 9		
Peri 0 Memory Mapped IO	peri_0_mmio_0	Memory-Mapped I/O-1.1
> Peri 1		
> Miscellaneous		

Device Configurator contd...

File Edit Settings View Help

PSE846GPS2DBZC4A CYW55513IU8G

Peripherals Analog Pins Clocks System **Memory** DMA

Enter filter text...

Memory Regions Memory Address View Peripheral Address View Read Me

Hide Detailed View

Resource	Name(s)	Personality
Memory	vres_0_memory_0	Memory Regions-2.0
Memory Region 0	vres_0_memory_region_data_0	memory_region_data-1.0
Memory Region 1	vres_0_memory_region_data_1	memory_region_data-1.0
Memory Region 2	vres_0_memory_region_data_2	memory_region_data-1.0
Memory Region 3	vres_0_memory_region_data_3	memory_region_data-1.0
Memory Region 4	vres_0_memory_region_data_4	memory_region_data-1.0
Memory Region 5	vres_0_memory_region_data_5	memory_region_data-1.0
Memory Region 6	vres_0_memory_region_data_6	memory_region_data-1.0
Memory Region 7	vres_0_memory_region_data_7	memory_region_data-1.0
Memory Region 8	vres_0_memory_region_data_8	memory_region_data-1.0
Memory Region 9	vres_0_memory_region_data_9	memory_region_data-1.0
Memory Region 10	vres_0_memory_region_data...	memory_region_data-1.0
Memory Region 11	vres_0_memory_region_data...	memory_region_data-1.0
Memory Region 12	vres_0_memory_region_data...	memory_region_data-1.0
Memory Region 13	vres_0_memory_region_data...	memory_region_data-1.0
Memory Region 14	vres_0_memory_region_data...	memory_region_data-1.0
Memory Region 15	vres_0_memory_region_data...	memory_region_data-1.0
Memory Region 16	vres_0_memory_region_data...	memory_region_data-1.0
Memory Region 17	vres_0_memory_region_data...	memory_region_data-1.0
Memory Region 18	vres_0_memory_region_data...	memory_region_data-1.0
Memory Region 19	vres_0_memory_region_data...	memory_region_data-1.0
Memory Region 20	vres_0_memory_region_data...	memory_region_data-1.0
Memory Region 21	vres_0_memory_region_data...	memory_region_data-1.0
Memory Region 22	vres_0_memory_region_data...	memory_region_data-1.0
Memory Region 23	vres_0_memory_region_data...	memory_region_data-1.0
Memory Region 24	vres_0_memory_region_data...	memory_region_data-1.0
+ Add: Memory Region 25 (26 of 64) vres_0_memory_region_data_25		

Instruction Tightly Coupled Memory (ITCM) 256 KB

Data Tightly Coupled Memory (DTCM) 256 KB

RRAM 512 KB

SRAM 1 MB

System SRAM (SoC MEM) 5 MB

Serial Memory Interface block 0, memory 1 (S25FS128S) 16 MB

23

Device Configurator contd...

File
Edit
Settings
View
Help

PSE846GPS2DBZC4A
CYW55513IUBG

Peripherals
Analog
Pins
Clocks
System
Memory
DMA

Enter filter text...

Memory Regions
Memory Address View
Peripheral Address View
Read Me

100%

Core
CM33
Enter search text...

Resource	Name(s)	Personality	Address	Regions	Locations
Memory	vres_0_memory_0	Memory Regions-2.0			
Memory Region 0	vres_0_memory_region_data_0	memory_region_data-1.0	0x00000000		
Memory Region 1	vres_0_memory_region_data_1	memory_region_data-1.0			
Memory Region 2	vres_0_memory_region_data_2	memory_region_data-1.0			
Memory Region 3	vres_0_memory_region_data_3	memory_region_data-1.0			
Memory Region 4	vres_0_memory_region_data_4	memory_region_data-1.0			
Memory Region 5	vres_0_memory_region_data_5	memory_region_data-1.0			
Memory Region 6	vres_0_memory_region_data_6	memory_region_data-1.0			
Memory Region 7	vres_0_memory_region_data_7	memory_region_data-1.0			
Memory Region 8	vres_0_memory_region_data_8	memory_region_data-1.0			
Memory Region 9	vres_0_memory_region_data_9	memory_region_data-1.0			
Memory Region 10	vres_0_memory_region_data...	memory_region_data-1.0			
Memory Region 11	vres_0_memory_region_data...	memory_region_data-1.0			
Memory Region 12	vres_0_memory_region_data...	memory_region_data-1.0			
Memory Region 13	vres_0_memory_region_data...	memory_region_data-1.0			
Memory Region 14	vres_0_memory_region_data...	memory_region_data-1.0			
Memory Region 15	vres_0_memory_region_data...	memory_region_data-1.0			
Memory Region 16	vres_0_memory_region_data...	memory_region_data-1.0			
Memory Region 17	vres_0_memory_region_data...	memory_region_data-1.0			
Memory Region 18	vres_0_memory_region_data...	memory_region_data-1.0			
Memory Region 19	vres_0_memory_region_data...	memory_region_data-1.0			
Memory Region 20	vres_0_memory_region_data...	memory_region_data-1.0			
Memory Region 21	vres_0_memory_region_data...	memory_region_data-1.0			
Memory Region 22	vres_0_memory_region_data...	memory_region_data-1.0			
Memory Region 23	vres_0_memory_region_data...	memory_region_data-1.0			
Memory Region 24	vres_0_memory_region_data...	memory_region_data-1.0			
+ Add: Memory Region 25 (26 of 64)	vres_0_memory_region_data_25				
			0x42000000	PERIO_MAIN (16 KB)	16 bit Divider 0 16 bit Divider 1 16 bit Divider 2 16 bit Divider 3 16.5 bit Divider 0 16.5 bit Divider 1 24.5 bit Divider 0 8 bit Divider 0 8 bit Divider 1 8 bit Divider 2 8 bit Divider 3 8 bit Divider 4 Peri 0 Peri 0 Clock Group 0 Peri 0 Clock Group 1 Peri 0 Clock Group 2 Peri 0 Clock Group 3 Peri 0 Clock Group 4 Peri 0 Clock Group 5 Peri 0 Clock Group 6 Peri 0 Clock Group 7 Peri 0 Clock Group 8 Peri 0 Clock Group 9
			0x42004000		
			0x42004010	PERIO_GRP0_GROUP (8 Bytes)	Peri 0 Memory Mapped IO
			0x42004018		

Device Configurator contd...

File Edit Settings View Help

PSE846GPS2DBZC4A CYW55513IU8G

Peripherals Analog Pins Clocks System **Memory** DMA

Enter filter text...

Memory Regions Memory Address View Peripheral Address View **Read Me**

Resource	Name(s)	Personality
Memory	vres_0_memory_0	Memory Regions-2.0
Memory Region 0	vres_0_memory_region_data_0	memory_region_data-1.0
Memory Region 1	vres_0_memory_region_data_1	memory_region_data-1.0
Memory Region 2	vres_0_memory_region_data_2	memory_region_data-1.0
Memory Region 3	vres_0_memory_region_data_3	memory_region_data-1.0
Memory Region 4	vres_0_memory_region_data_4	memory_region_data-1.0
Memory Region 5	vres_0_memory_region_data_5	memory_region_data-1.0
Memory Region 6	vres_0_memory_region_data_6	memory_region_data-1.0
Memory Region 7	vres_0_memory_region_data_7	memory_region_data-1.0
Memory Region 8	vres_0_memory_region_data_8	memory_region_data-1.0
Memory Region 9	vres_0_memory_region_data_9	memory_region_data-1.0
Memory Region 10	vres_0_memory_region_data...	memory_region_data-1.0
Memory Region 11	vres_0_memory_region_data...	memory_region_data-1.0
Memory Region 12	vres_0_memory_region_data...	memory_region_data-1.0
Memory Region 13	vres_0_memory_region_data...	memory_region_data-1.0
Memory Region 14	vres_0_memory_region_data...	memory_region_data-1.0
Memory Region 15	vres_0_memory_region_data...	memory_region_data-1.0
Memory Region 16	vres_0_memory_region_data...	memory_region_data-1.0
Memory Region 17	vres_0_memory_region_data...	memory_region_data-1.0
Memory Region 18	vres_0_memory_region_data...	memory_region_data-1.0
Memory Region 19	vres_0_memory_region_data...	memory_region_data-1.0
Memory Region 20	vres_0_memory_region_data...	memory_region_data-1.0
Memory Region 21	vres_0_memory_region_data...	memory_region_data-1.0
Memory Region 22	vres_0_memory_region_data...	memory_region_data-1.0
Memory Region 23	vres_0_memory_region_data...	memory_region_data-1.0
Memory Region 24	vres_0_memory_region_data...	memory_region_data-1.0
+ Add: Memory Region 25 (26 of 64)	vres_0_memory_region_data_25	

Overview

The PSOC Edge MCU features two CPU subsystems. The CM33 CPU comes with a 32KB instruction cache while the CM55 CPU comes with a 32KB instruction cache along with a 32KB data cache. The MCU has two 512KB SRAM sections in the low-power domain and 5120KB of System SRAM (SoCMEM) in the high-performance domain. Additionally, it includes 512KB of RRAM for non-volatile storage, primarily for code and selective application data. The CM55 core's performance is further enhanced by 256KB of tightly coupled memory for data and instructions. For more information, refer to AN239774.

- [How to read the build output and associate it with the regions defined in the tool](#)
- [How to define a new region for some data buffer variable](#)
- [How to define a new region for the code and then assign code to the particular region using pragma directives](#)
- [How to assign a function or variable into a custom memory section](#)
- [How to modify the linker to include a specific file into a custom memory section](#)
- [How to modify the linker to include an entire middleware library into a custom memory section](#)
- [How to re-balance the memory between the cores based on the build output](#)
- [How to switch between internal and external memory](#)

Instruction Tightly Coupled Memory (ITCM) (256 KB)

ITCM stores program instructions requiring low latency and minimal contention for access, often housing critical parts of the program like interrupt service routines (ISRs), time-critical control loops, or frequently used functions. ITCM instructions do not pass through the cache and do not impose any restriction on wait states. This allows for direct and deterministic access to instructions from this TCM, improving execution speed for critical, or time-sensitive code. Accessing instructions from ITCM typically incurs fewer or zero wait states compared to accessing instructions from slower memory regions like flash memory. This reduction in wait states enhances the real-time performance and responsiveness of the MCU.

Address	Cores	Executable	Writable	Readable	Secure
CM55_ITCM_CM33 0x48000000-0x4803FFFF	CM33 CM33 (secure)	✓	✓	✓	
CM55_ITCM_CM33_S 0x58000000-0x5803FFFF	CM33 (secure)	✓	✓	✓	✓
CM55_ITCM_INTERNAL 0x00000000-0x0003FFFF	CM55	✓	✓	✓	

Data Tightly Coupled Memory (DTCM) (256 KB)

Device Configurator contd...

File

Edit

Settings

View

Help

PSE846GPS2DBZC4A

CYW55513IUBG

Peripherals

Analog

Pins

Clocks

System

Memory

DMA

Enter filter text...

Resource	Name(s)	Personality
> AXIDMAC 0		
> DMA		
> DMA DataWire 0		
> DMA DataWire 1		
☐ DMA DataWire 1: Channel 0	m33syscpuss_0_dw1_0_chan_0	
☐ DMA DataWire 1: Channel 1	m33syscpuss_0_dw1_0_chan_1	
☐ DMA DataWire 1: Channel 2	m33syscpuss_0_dw1_0_chan_2	
☐ DMA DataWire 1: Channel 3	m33syscpuss_0_dw1_0_chan_3	
☐ DMA DataWire 1: Channel 4	m33syscpuss_0_dw1_0_chan_4	
☐ DMA DataWire 1: Channel 5	m33syscpuss_0_dw1_0_chan_5	
☐ DMA DataWire 1: Channel 6	m33syscpuss_0_dw1_0_chan_6	
☐ DMA DataWire 1: Channel 7	m33syscpuss_0_dw1_0_chan_7	
☐ DMA DataWire 1: Channel 8	m33syscpuss_0_dw1_0_chan_8	
☐ DMA DataWire 1: Channel 9	m33syscpuss_0_dw1_0_chan_9	
☐ DMA DataWire 1: Channel 10	m33syscpuss_0_dw1_0_chan...	
☐ DMA DataWire 1: Channel 11	m33syscpuss_0_dw1_0_chan...	
☐ DMA DataWire 1: Channel 12	m33syscpuss_0_dw1_0_chan...	
☐ DMA DataWire 1: Channel 13	m33syscpuss_0_dw1_0_chan...	
☐ DMA DataWire 1: Channel 14	m33syscpuss_0_dw1_0_chan...	
☐ DMA DataWire 1: Channel 15	m33syscpuss_0_dw1_0_chan...	

QSPI Configurator

File
Edit
Settings
View
Help



Target Device Family:
PSE846GOS4DBZC4
Flashloader Directory:

Program Chunk Size:
4 KB
Erase Chunk Size:
256 KB

QSPI Instance	Slave Slot	Memory Part Number	Configuration	Data Select	Memory Mapped	Pair with Slot	Start Address	Size	End Address	Write Enable	Config Data in Flash	Encrypt	Merge XIP Transactions
0	0	S25FS128S	Hybrid at bottom, 256 KB sectors elsewh	Quad SPI-Data[0:3]	☒	None	0x60000000	0x1000000	0x60FFFFFF	☒	☐	☐	Within 256 Cycles
0	1	Not Used		Octal SPI-Data[0:7]	☒	None	0x61000000	0x1000000	0x61FFFFFF	☒	☐	☐	Within 256 Cycles
0	2	Not Used		Octal SPI-Data[0:7]	☐	None	0x64000000	0x4000000	0x67FFFFFF	☐	☐	☐	Disabled
0	3	Not Used		SPI-MOSI:MISO Data[0:1]	☐	None	0x60000000	0x10000	0x6000FFFF	☐	☐	☐	Disabled
1	0	Not Used		Octal SPI-Data[0:7]	☒	None	0x64000000	0x4000000	0x67FFFFFF	☐	☐	☐	Within 256 Cycles
1	1	Not Used		SPI-MOSI:MISO Data[0:1]	☐	None	0x60000000	0x10000	0x6000FFFF	☐	☐	☐	Disabled
1	2	Not Used		Dual SPI-Data[0:1]	☐	None	0x60000000	0x10000	0x6000FFFF	☐	☐	☐	Disabled
1	3	Not Used		SPI-MOSI:MISO Data[0:1]	☐	None	0x60000000	0x10000	0x6000FFFF	☐	☐	☐	Disabled

Location:
C:/Users/RamakrishnaP/.modustoolbox/global/device-db/develop/devices/memory/S25FS128S-256KB-hybridbottom/qspi_1.0/view.xml

Memory Part Number:
S25FS128S
HyperBus Device:
☐

Configuration:
Hybrid at bottom, 256 KB sectors elsewhere
Configuration Is Default:
☒

Description:
16MBytes 1.8V serial Flash memory

General
Commands
Hybrid Erase Sectors

Registers
Status Register Busy (WIP) Mask:
0x01
Status Register Quad Enable (QE) Mask:
0x02
Status Register Octal Enable (OE) Mask:
0x00

Timings
Max Erase Time:
2.9
s
Max Chip Erase Time:
180
s
Max Program Time (us):
2000

Octal DDR Enable Sequence
Command Sequence 1:
NA
Command Sequence 2:
NA

SMART I/O Configurator

File
Settings
View
Help

Routing
LUT 1
LUT 2
LUT 3

Port: Smart I/O 11
Clock: Peripheral clock divider (Active)
Clock divider: 24.5 bit Divider 0 clk (CYBSP_SMART_IO_CLK_DIV) [USED]
Show Routing Matrix
Clear

Chip 7 Bypass

Chip 6 Bypass

Chip 5 Bypass

Chip 4 Bypass

Chip 3 Input(Async)
TCPWM[0] Group[1] 16-bit Counter 6 pwm_n (CYBSP_SMARTIO_PWM) [USED]

Chip 2 Bypass

Chip 1 Bypass

Chip 0 Bypass

0	1	2
LUT 0	LUT 1	LUT 2
None	LUT2	LUT2
None	LUT1	LUT2
None	LUT1	LUT2

TR 0: rst

TR 1: en

TR 2: [UNUSED]

Constant 0

Constant 0

Constant 0

I/O 7 Bypass

I/O 6 Bypass

I/O 5 Bypass

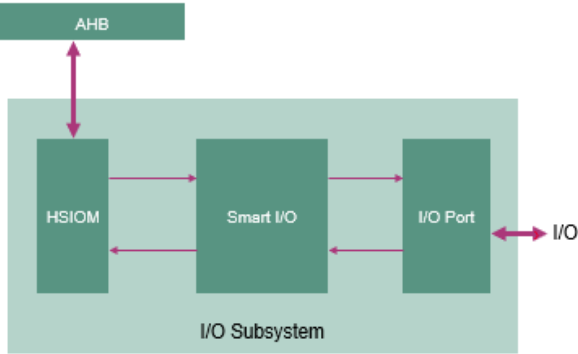
I/O 4 Bypass

I/O 3 Output

I/O 2 Bypass

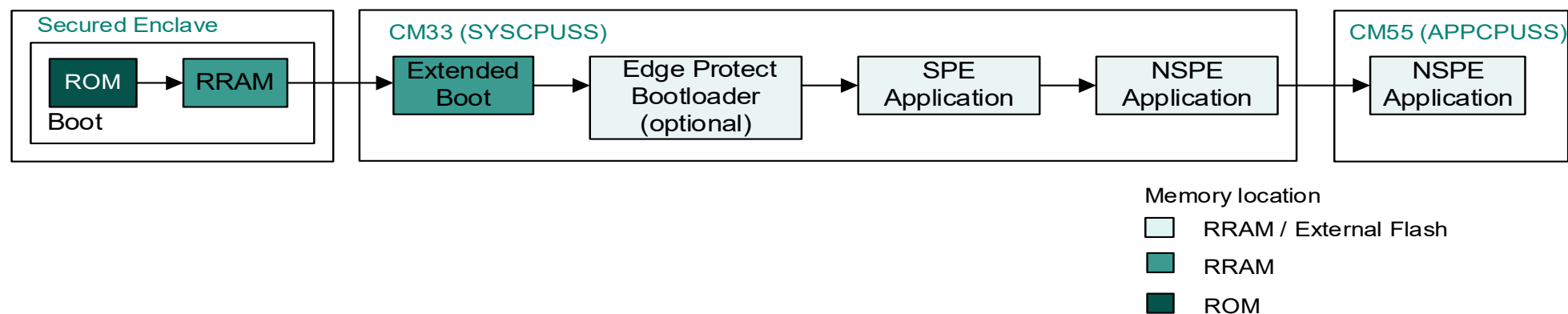
I/O 1 Bypass

I/O 0 Bypass



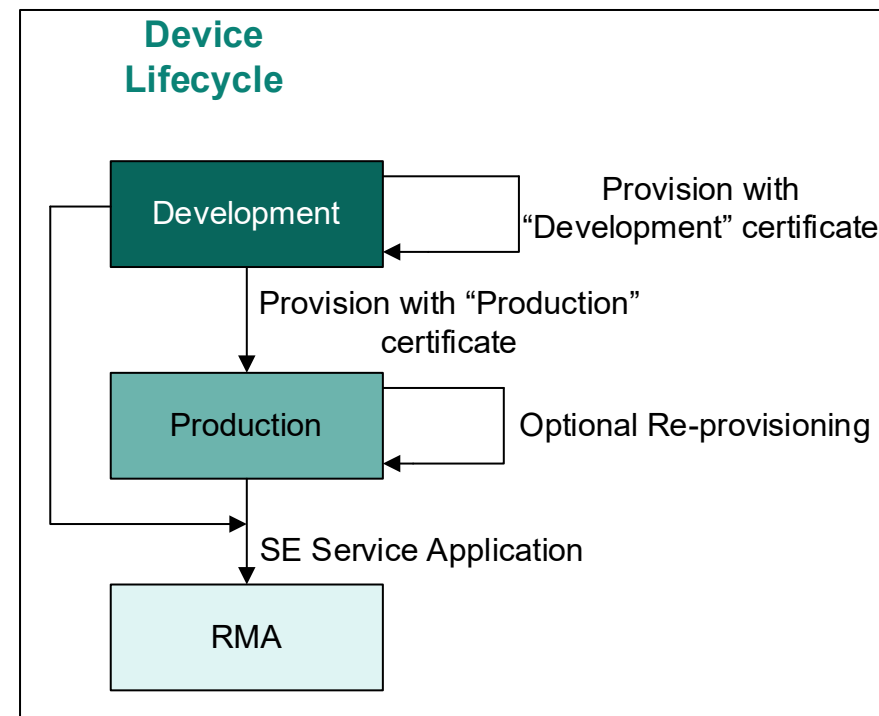
Application Booting sequence

- Boot starts from Secured Enclave as the Root of Trust
- Off-the-shelf extended boot software
 - Boot from RRAM or external flash
 - Alternate serial interface for recovery boot
 - SRAM loading for enhanced security
- Optional MCUboot based Edge Protect Bootloader
- One-time revocable OEM RoT Key



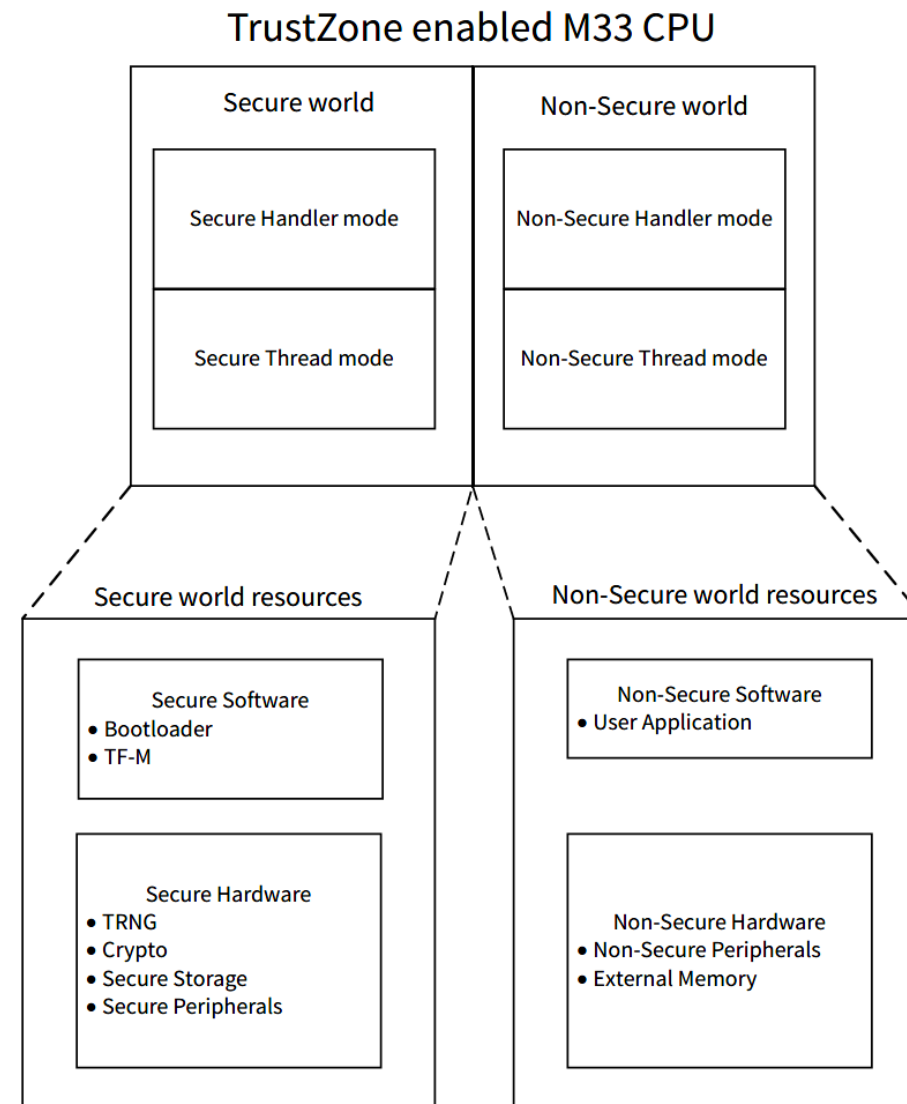
Device lifecycle and provisioning

- Development
 - Shipped from Infineon
 - Provision with OEM certificate and policy
 - For OEM development; until production
- Production
 - One-time provisioning with “Production” certificate
 - Re-provisioning if allowed by policy
- RMA
 - Transition from Development or Production
 - Performed by Infineon signed SE Service Application
 - Devices in RMA mode can be shipped to Infineon for field failure analysis

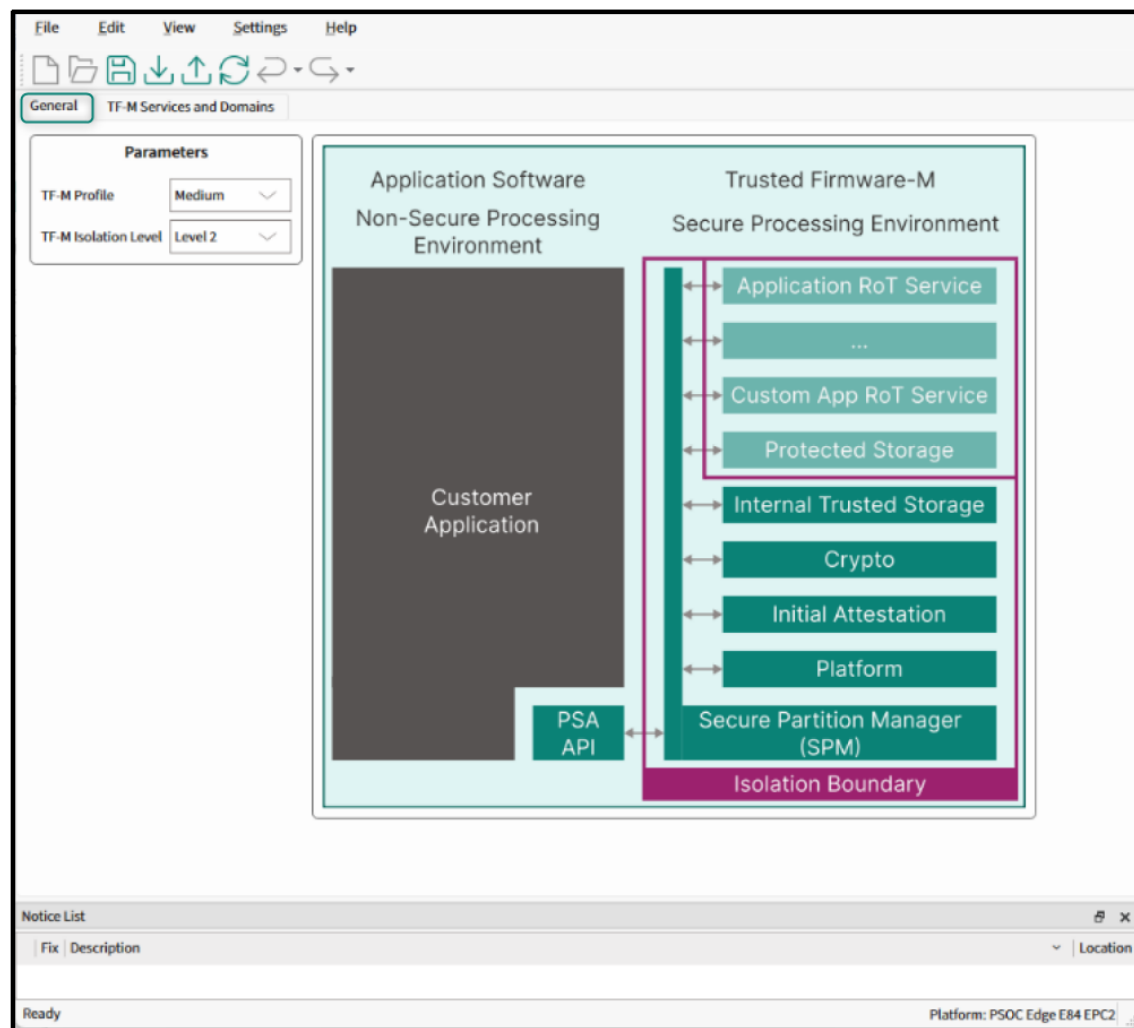


Secure and Non-Secure Operating Modes

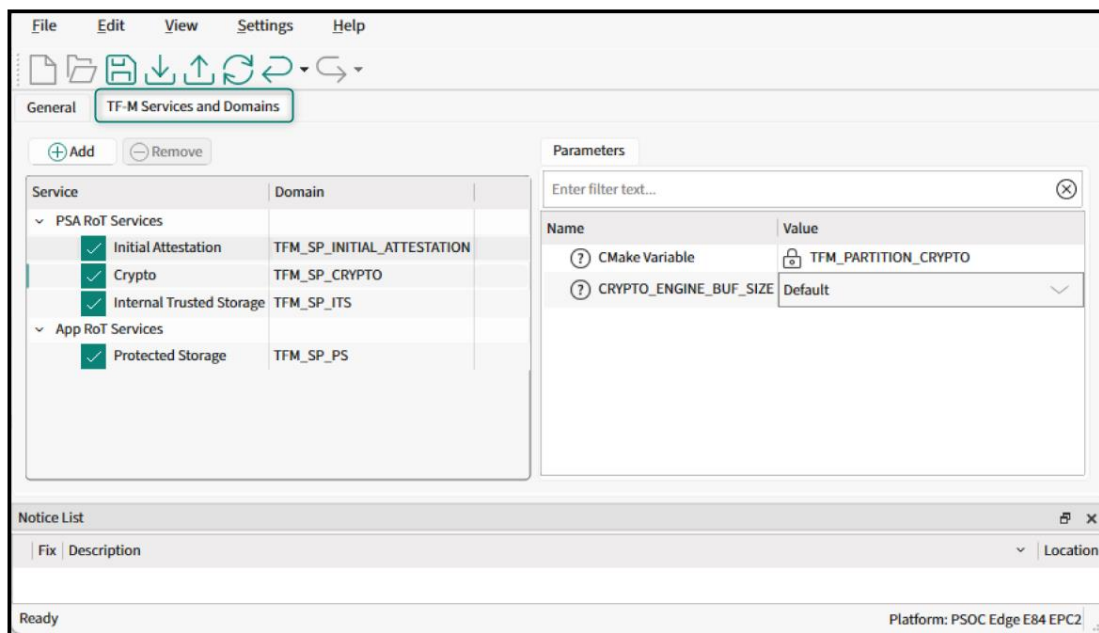
- Arm® TrustZone- enabled Cortex® -M33 CPU
- Secure and non-secure CPU states
- Attribution units - Secure Attribution Unit (SAU) and Implementation Defined Attribution Unit (IDAU)
- Banked resources between secure and non-secure states
- Secure and non-secure
 - aliasing of the memory map
 - interrupts
 - peripherals
 - debug



Edge Protect Configurator

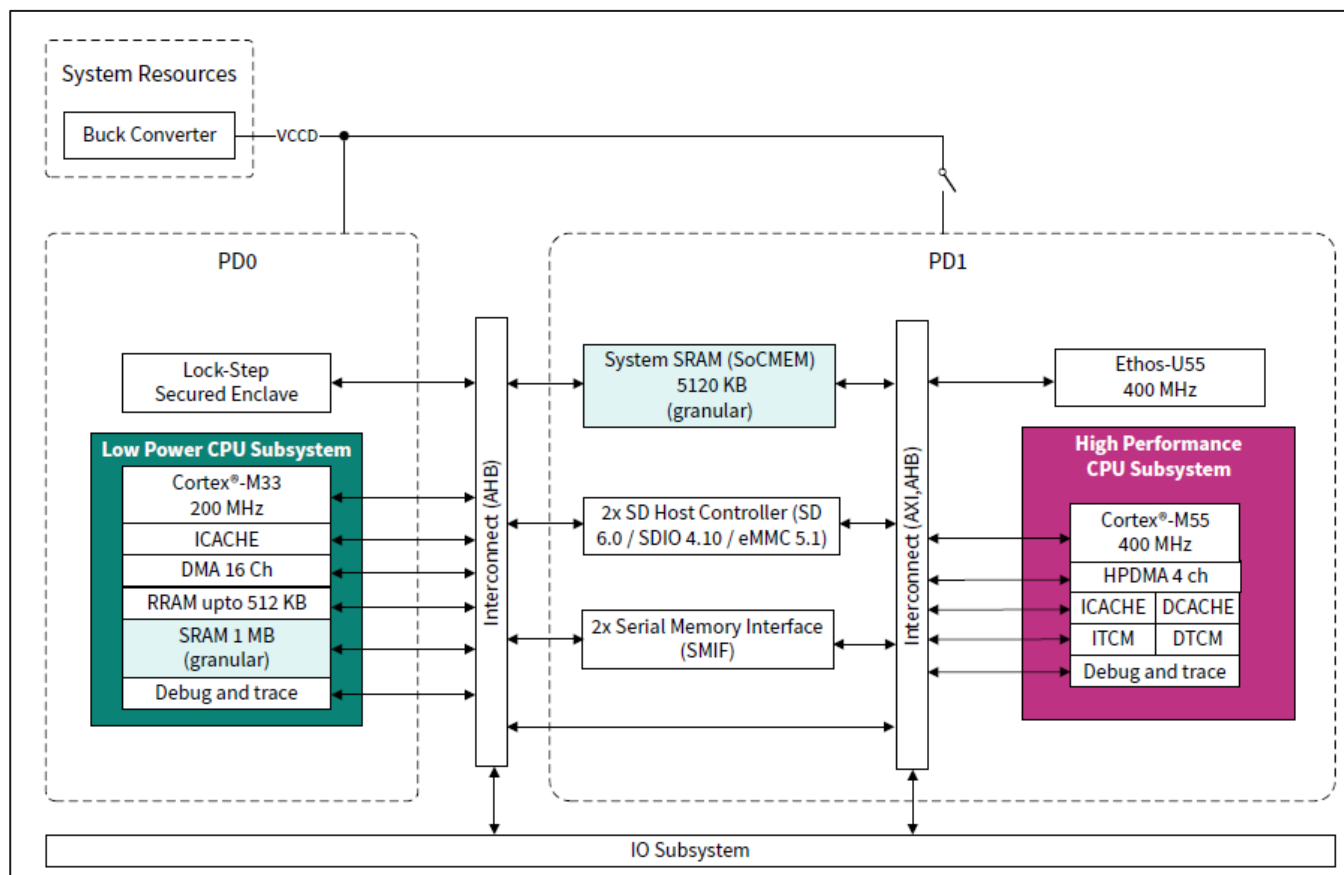


General tab



TF-M Services and Domains tab

PSOC™ Edge memory architecture



- The PSOC™ Edge features two power domains:
 - PD0: Always-on low-power domain.
 - PD1: High-performance domain.
- Power Source and Operating voltage:
 - Both PD0 and PD1 are powered by the internal buck output and operate at same voltage level.
- Independent Control and flexibility:
 - PD1 can be independently switched off without affecting PD0.
 - PD1 can be reactivated as necessary for performance needs.
- Power Conservation:
 - When high-performance peripherals are not needed, PD1 can be turned off to save power.

Linker Scripts

- Four top-level linker files
 - Bootloader
 - CM33 Secure
 - CM33 Non-secure
 - CM55

- Three generated memory *include* files
 - CM33 Secure
 - CM33 Non-secure
 - CM55

- Generated memory *include* files come from the Device Configurator

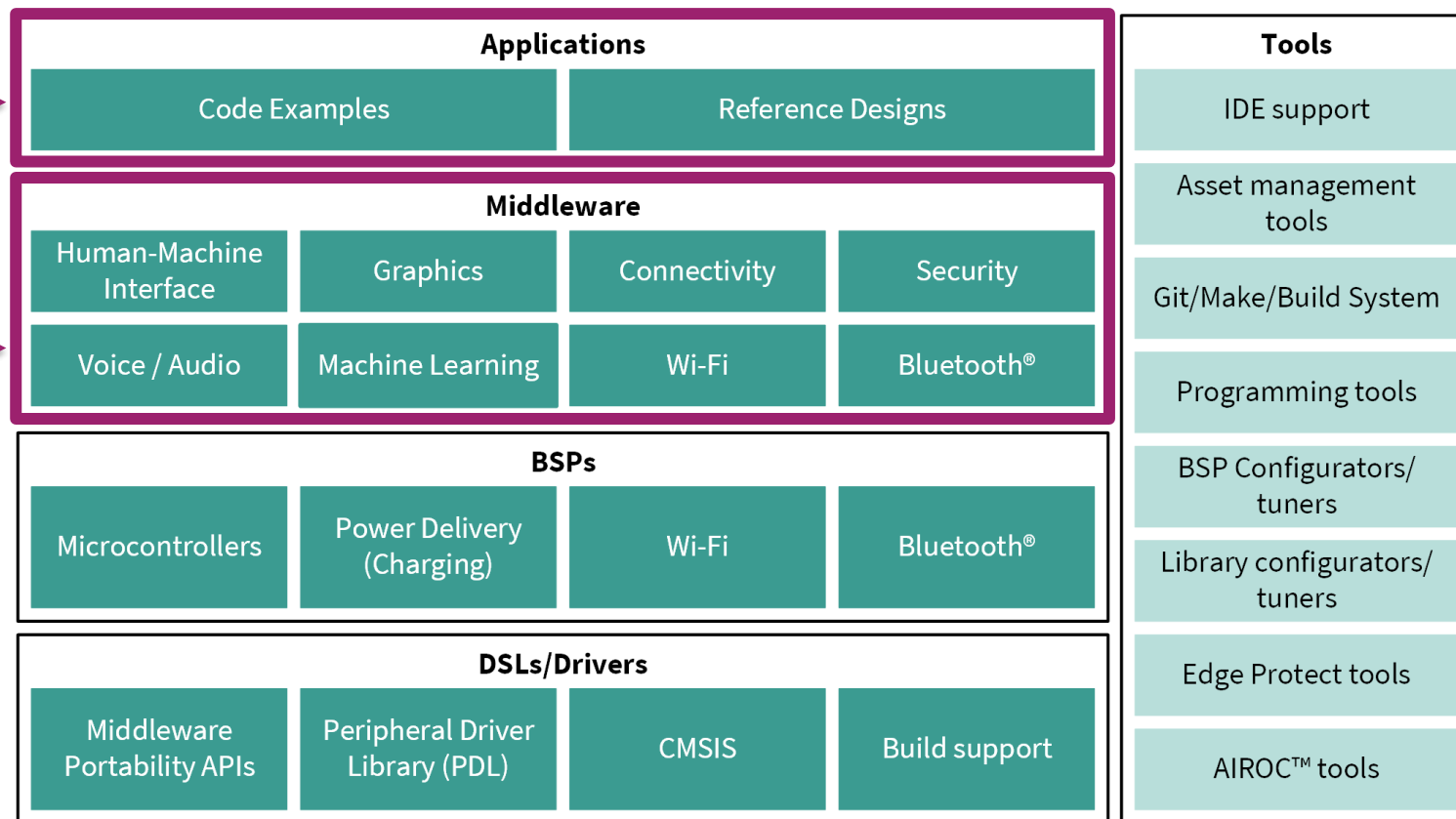
- **New user defined memory regions require direct modification to top-level linker files to leverage generated content**



Firmware/application development :Software Resources

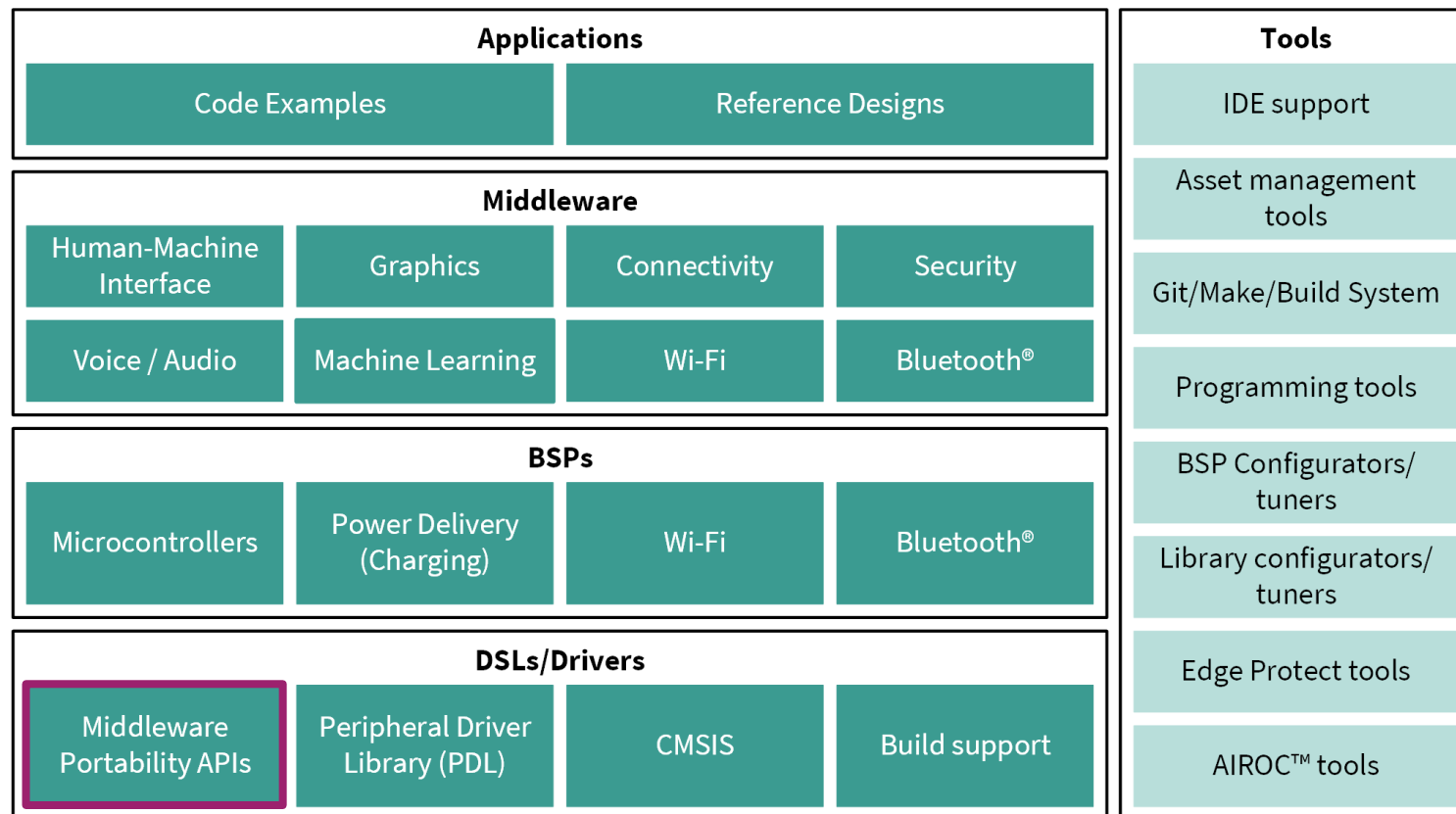
- Applications
 - Code examples
 - Reference Designs

- Middleware Libraries



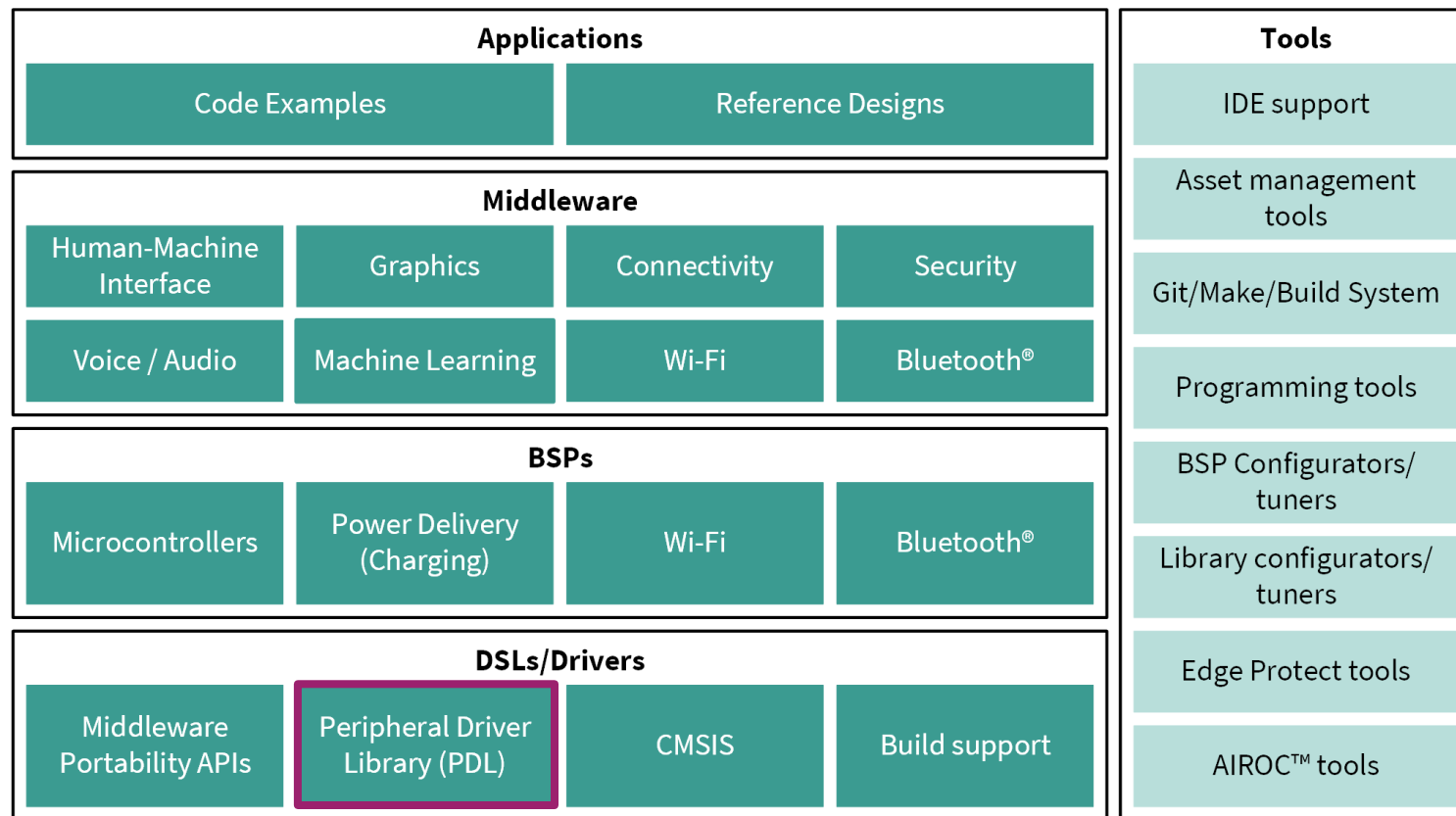
Firmware/application development : Software Resources

- Middleware Portability APIs
 - High-level APIs built on top of PDL
 - Hardware generic, so more portable
 - Architected to enable Infineon maintained middleware to work across a wider range of devices
 - Replacement of the former ModusToolbox™ HAL layer



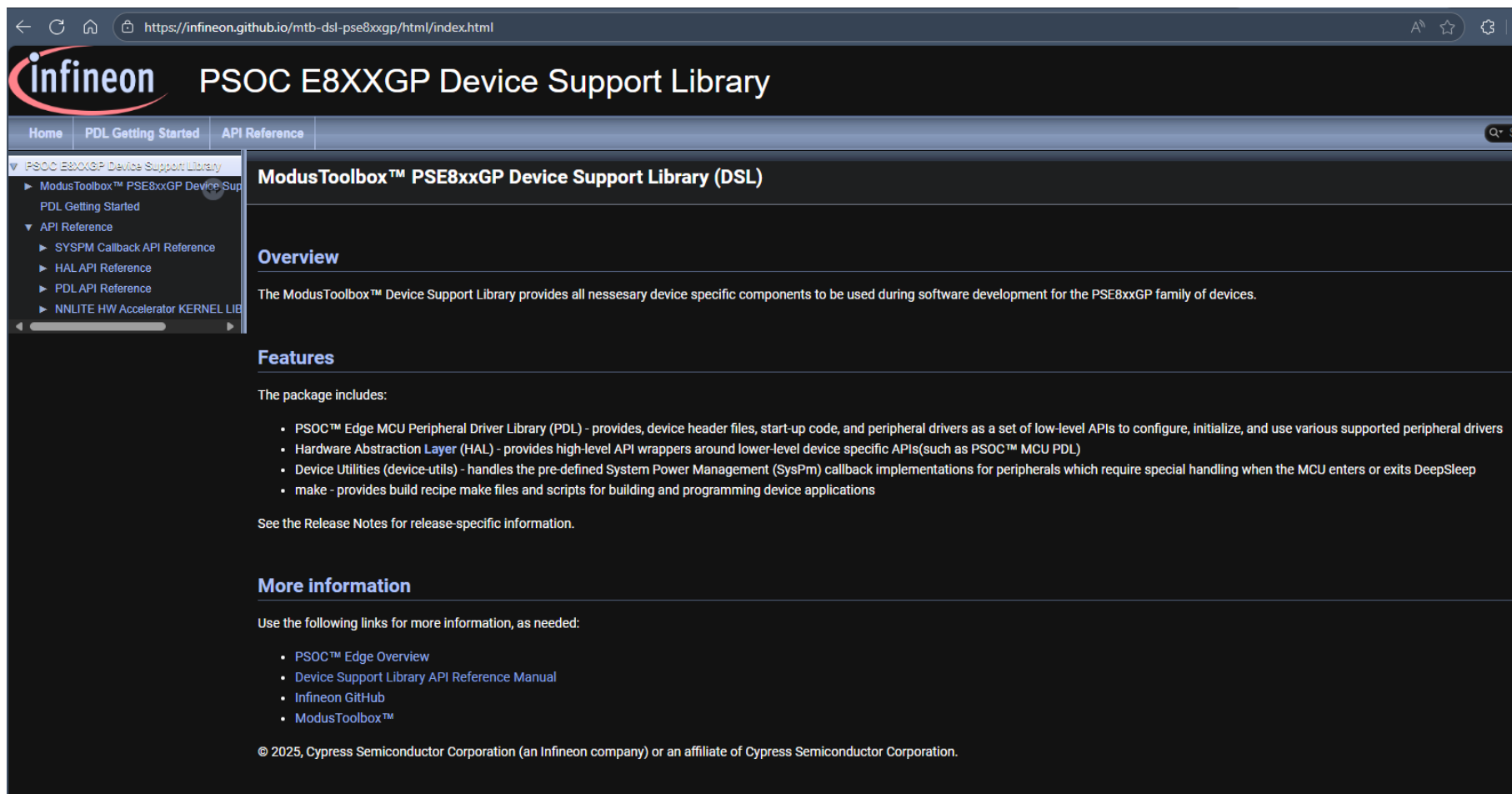
Firmware/application development : Software Resources

- Peripheral Driver Library (PDL)
 - Set of low-level APIs
 - Hardware specific so less portable



Firmware/application development : Software Resources

<https://infineon.github.io/mtb-dsl-pse8xxgp/html/index.html>



infineon PSOC E8XXGP Device Support Library

Home PDL Getting Started API Reference

PSOC E8XXGP Device Support Library

- ModusToolbox™ PSE8xxGP Device Sup
- PDL Getting Started
- API Reference
 - SYSPM Callback API Reference
 - HAL API Reference
 - PDL API Reference
 - NNLITE HW Accelerator KERNEL LIB

ModusToolbox™ PSE8xxGP Device Support Library (DSL)

Overview

The ModusToolbox™ Device Support Library provides all nessesary device specific components to be used during software development for the PSE8xxGP family of devices.

Features

The package includes:

- PSOC™ Edge MCU Peripheral Driver Library (PDL) - provides, device header files, start-up code, and peripheral drivers as a set of low-level APIs to configure, initialize, and use various supported peripheral drivers
- Hardware Abstraction Layer (HAL) - provides high-level API wrappers around lower-level device specific APIs(such as PSOC™ MCU PDL)
- Device Utilities (device-utils) - handles the pre-defined System Power Management (SysPm) callback implementations for peripherals which require special handling when the MCU enters or exits DeepSleep
- make - provides build recipe make files and scripts for building and programming device applications

See the Release Notes for release-specific information.

More information

Use the following links for more information, as needed:

- PSOC™ Edge Overview
- Device Support Library API Reference Manual
- Infineon GitHub
- ModusToolbox™

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FreeRTOS support with ModusToolbox™

The screenshot shows the Eclipse IDE interface with the ModusToolbox™ extension. The 'Project Explorer' on the left lists the project 'PSOC_Edge_Hello_World'. The 'Library Manager 2.50' window is open, displaying a list of libraries. The 'freertos' library is selected, and its version '10.6.201 release' is highlighted. The 'Add Library - Library Manager 2.50' dialog is also open, showing the 'freertos' library details and a 'License Disclaimer' section. The 'OK' button is highlighted.

Library Manager 2.50

Application Directory: C:/Workspace/PSOC_Edge_Hello_World

Enter filter text...

Libraries

- BSPs
 - APP_KIT_PSE84_EVAL_EPC2 (AC)
- proj_cm33_ns Libraries
 - async-transfer
 - bt-fw-ixf-cyw55500a1
 - cmsis
 - core-lib
 - core-make
 - device-db
 - mtb-dsl-pse8xxgp
 - mtb-ipc
 - mtb-srf
 - retarget-io
 - se-rt-services-utils
- proj_cm33_s Libraries
 - async-transfer
 - bt-fw-ixf-cyw55500a1
 - cmsis
 - core-lib

Add Library - Library Manager 2.50

Target Project: proj_cm55

Name	Version
☒ freertos	10.6.201 release
☐ freertos-posix	1.0.1 release
☐ Ethernet	
☐ ethernet-core-freertos-lwip-mbedtls	3.0.0 release
☐ Middleware	
☐ aws-iot-device-sdk-embedded-C	v4_beta release
☐ lwip-freertos-integration	1.2.0 release
☐ lwip-network-interface-integration	1.7.0 release
☐ Utilities	
☐ abstraction-rtos	1.12.0 release
☐ cliib-support	1.8.0 release
☐ Wi-Fi	
☐ wifi-core-freertos-lwip-mbedtls	3.1.0 release
☐ wifi-mw-core	3.4.0 release

freertos

A FreeRTOS is a small operating system for microcontrollers. A FreeRTOS kernel is distributed as standard C source files with configuration header file.

License Disclaimer:
Adding this library will download and add FreeRTOS to your project. It is your responsibility to understand and accept the FreeRTOS license.

Additional Information:

- FreeRTOS kernel README.md
- FreeRTOS kernel RELEASE.md
- FreeRTOS home page

Version details: 10.6.201 release

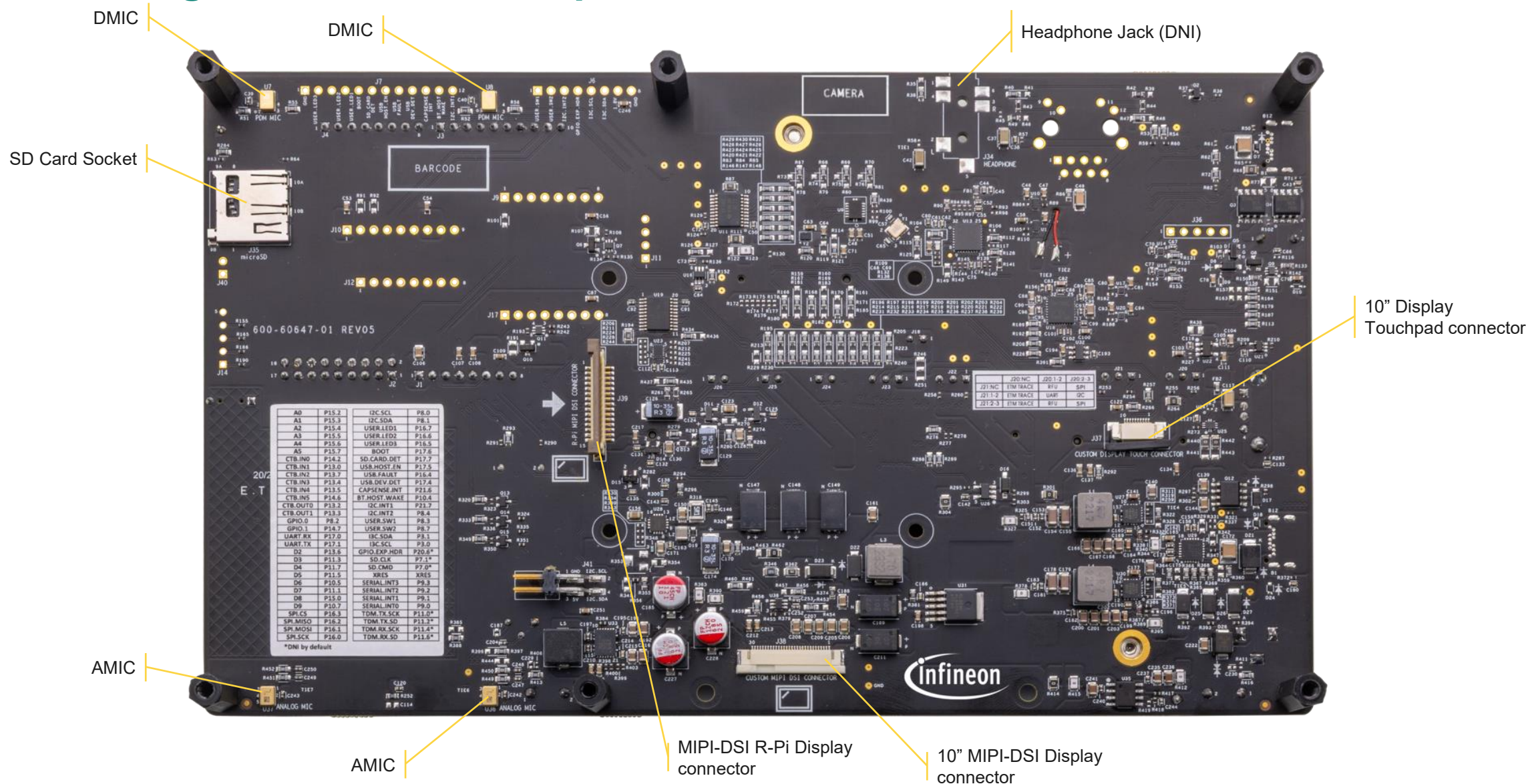
OK **Cancel**

Update **Close**

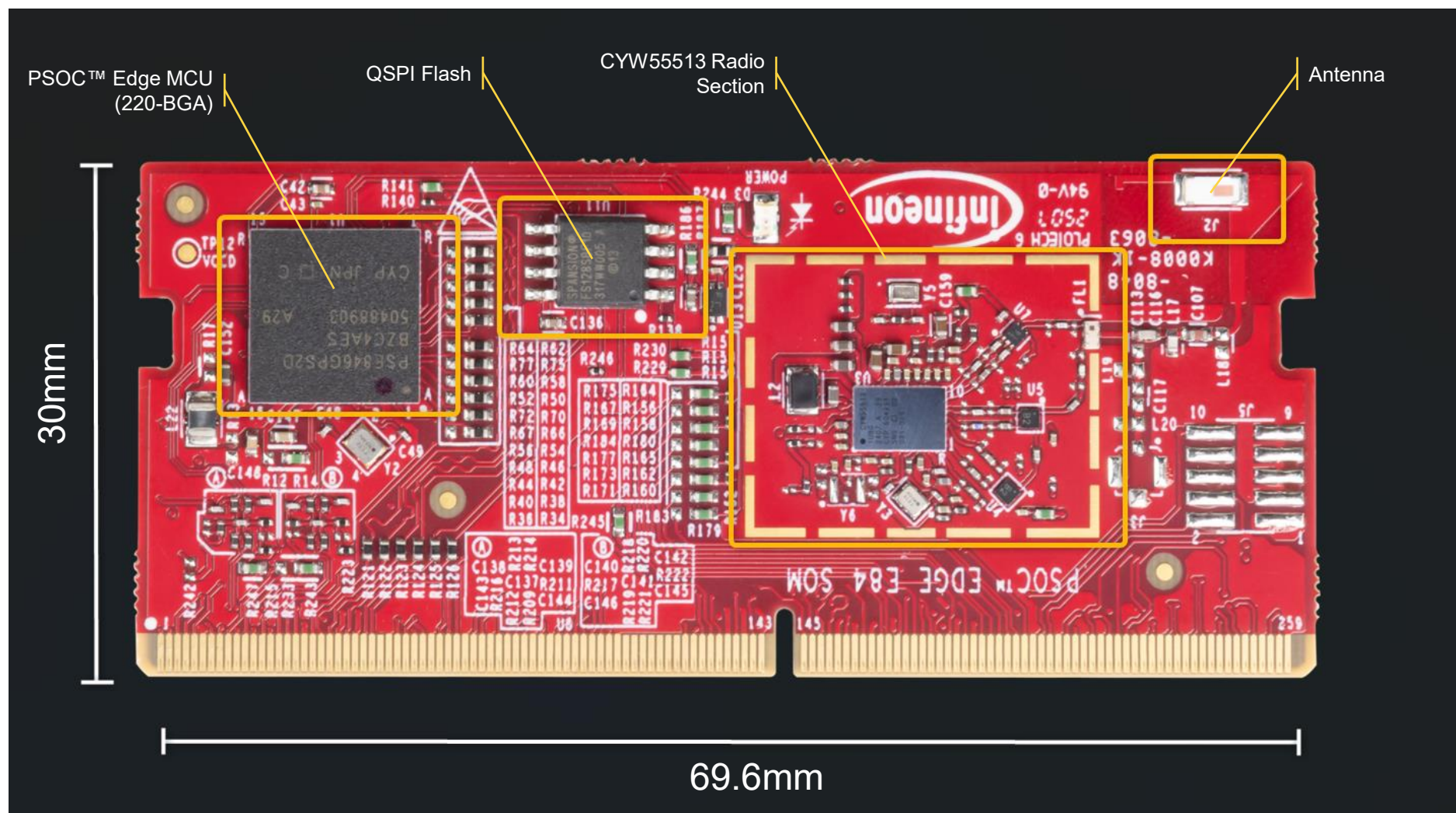
PSOC™ Edge E84 MCU development kit – Top View



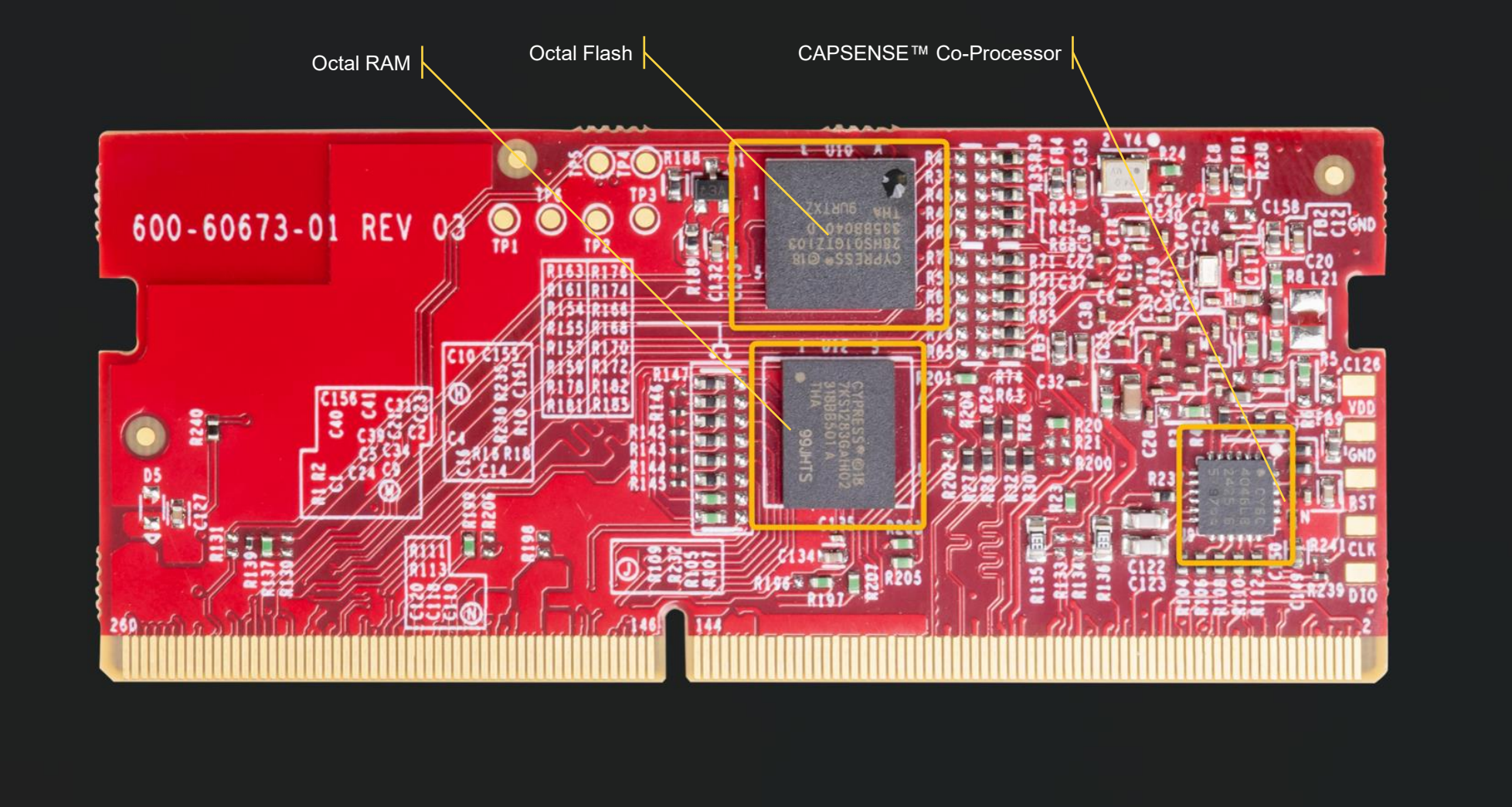
PSOC™ Edge E84 MCU development kit – Back View



PSOC™ Edge E84 MCU SOM – Top View



PSOC™ Edge E84 MCU SOM – Bottom View



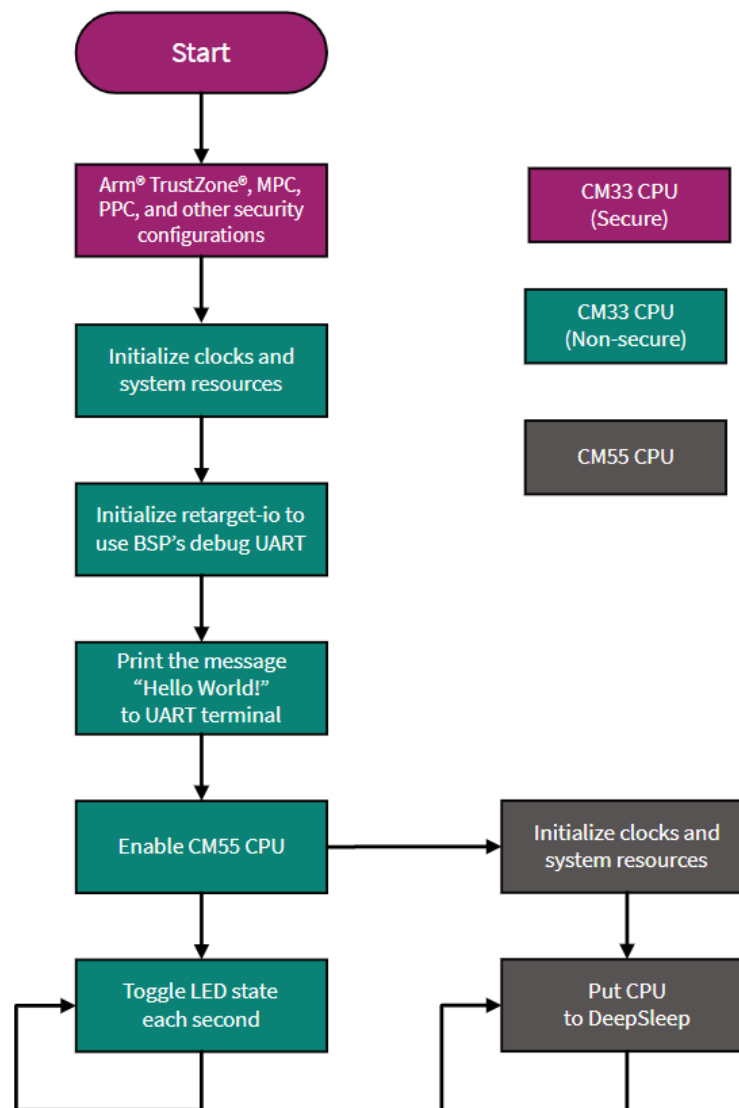
Programming/debugging

- Programming and Debugging PSOC™ Edge E84 MCU.
 - Comes with a KitProg3 onboard programmer and debugger.
 - Supports CMSIS-DAP.
 - Supports OpenOCD using a GDB server and the J-Link debug probe.
 - Debug environment is IDE specific – refer to ModusToolbox™ documentation.

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1	Development ecosystem for PSOC™ Edge E84	3
2	Code Example #1: Hello World + Blinky	45
3	Code Example #2: IPC Semaphores using PDL API	49

Description of Hello World + Blinky example



Using Device Configurator

PSE846GPS2DBZC4A CYW55513IUBG

Peripherals Analog Pins Clocks System Memory DMA

Enter filter text...

Resource	Name(s)	Personality
> Port 11		
> Port 12		
> Port 13		
> Port 14		
> Port 15		
▼ Port 16		
☒ P16[0]	CYBSP_SPI_CLK	Pin-3.0 ▼
☒ P16[1]	CYBSP_SPI_MOSI	Pin-3.0 ▼
☒ P16[2]	CYBSP_SPI_MISO	Pin-3.0 ▼
☒ P16[3]	CYBSP_SPI_CS	Pin-3.0 ▼
☐ P16[4]	CYBSP_USB_FAULT	
☒ P16[5]	CYBSP_USER_LED3, CYBSP_LED_BLUE	Pin-3.0 ▼
☒ P16[6]	CYBSP_USER_LED2, CYBSP_LED_GREEN	Pin-3.0 ▼
☒ P16[7]	TRAINING_LED, CYBSP_USER_LED1, CYBSP_USER_LED, CYBSP_LED_RED	Pin-3.0 ▼
> Port 17		
> Port 18		
> Port 19		
> Port 20		
> Port 21		

```

README.md  H cycfg_pins.h X
#define CYBSP_LED_GREEN_INIT_DRIVESTATE CYBSP_USER_LED2_INIT_DRIVESTATE
#ifndef ioss_0_port_16_pin_6_HSIOM
    #define ioss_0_port_16_pin_6_HSIOM HSIOM_SEL_GPIO
#endif
#define CYBSP_USER_LED2_HSIOM ioss_0_port_16_pin_6_HSIOM
#define CYBSP_LED_GREEN_HSIOM CYBSP_USER_LED2_HSIOM
#define CYBSP_USER_LED2_IRQ ioss_interrupts_gpio_16_IRQn
#define CYBSP_LED_GREEN_IRQ CYBSP_USER_LED2_IRQ
#define TRAINING_LED_ENABLED 1U
#define CYBSP_USER_LED1_ENABLED TRAINING_LED_ENABLED
#define CYBSP_USER_LED_ENABLED TRAINING_LED_ENABLED
#define CYBSP_LED_RED_ENABLED TRAINING_LED_ENABLED
#define TRAINING_LED_PORT GPIO_PRT16
#define CYBSP_USER_LED1_PORT TRAINING_LED_PORT
#define CYBSP_USER_LED_PORT TRAINING_LED_PORT
#define CYBSP_LED_RED_PORT TRAINING_LED_PORT
#define TRAINING_LED_PORT_NUM 16U
#define CYBSP_USER_LED1_PORT_NUM TRAINING_LED_PORT_NUM
#define CYBSP_USER_LED_PORT_NUM TRAINING_LED_PORT_NUM
#define CYBSP_LED_RED_PORT_NUM TRAINING_LED_PORT_NUM
#define TRAINING_LED_PIN 7U
#define CYBSP_USER_LED1_PIN TRAINING_LED_PIN
#define CYBSP_USER_LED_PIN TRAINING_LED_PIN
#define CYBSP_LED_RED_PIN TRAINING_LED_PIN
#define TRAINING_LED_NUM 7U
#define CYBSP_USER_LED1_NUM TRAINING_LED_NUM

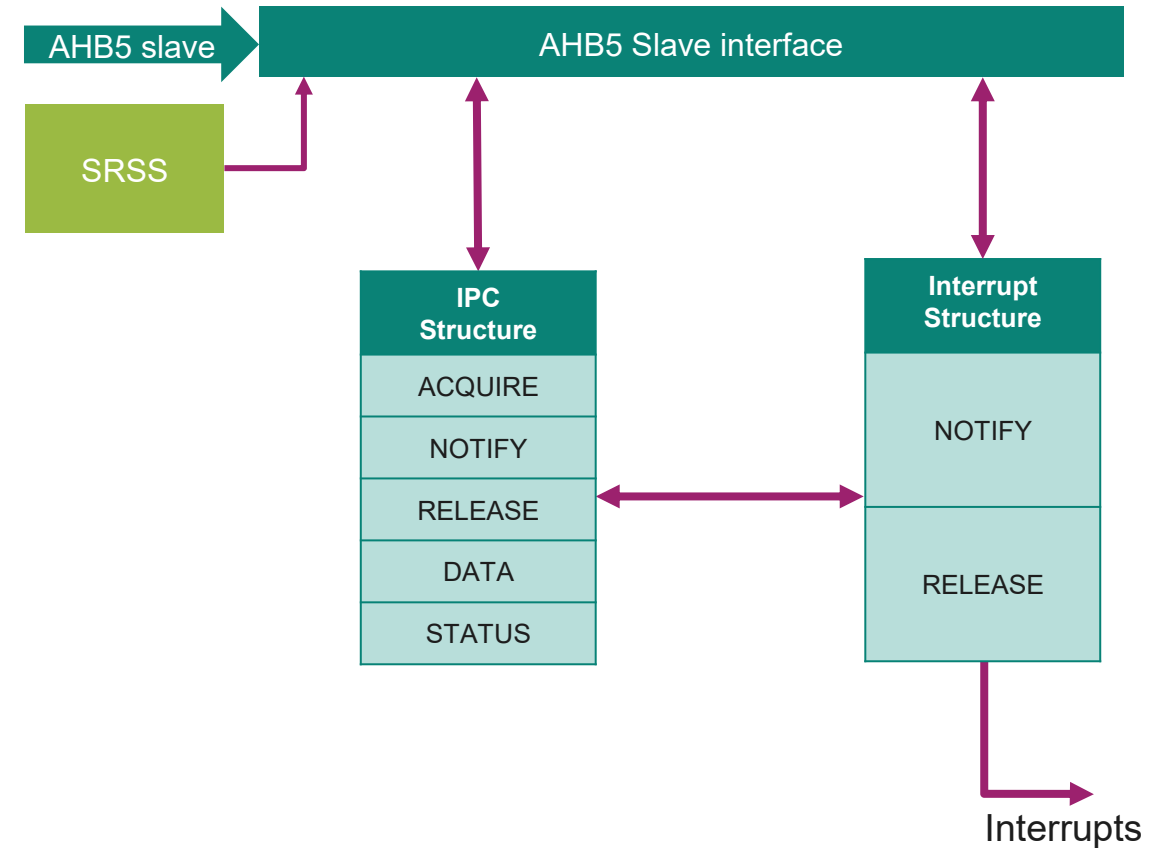
```

Table of contents

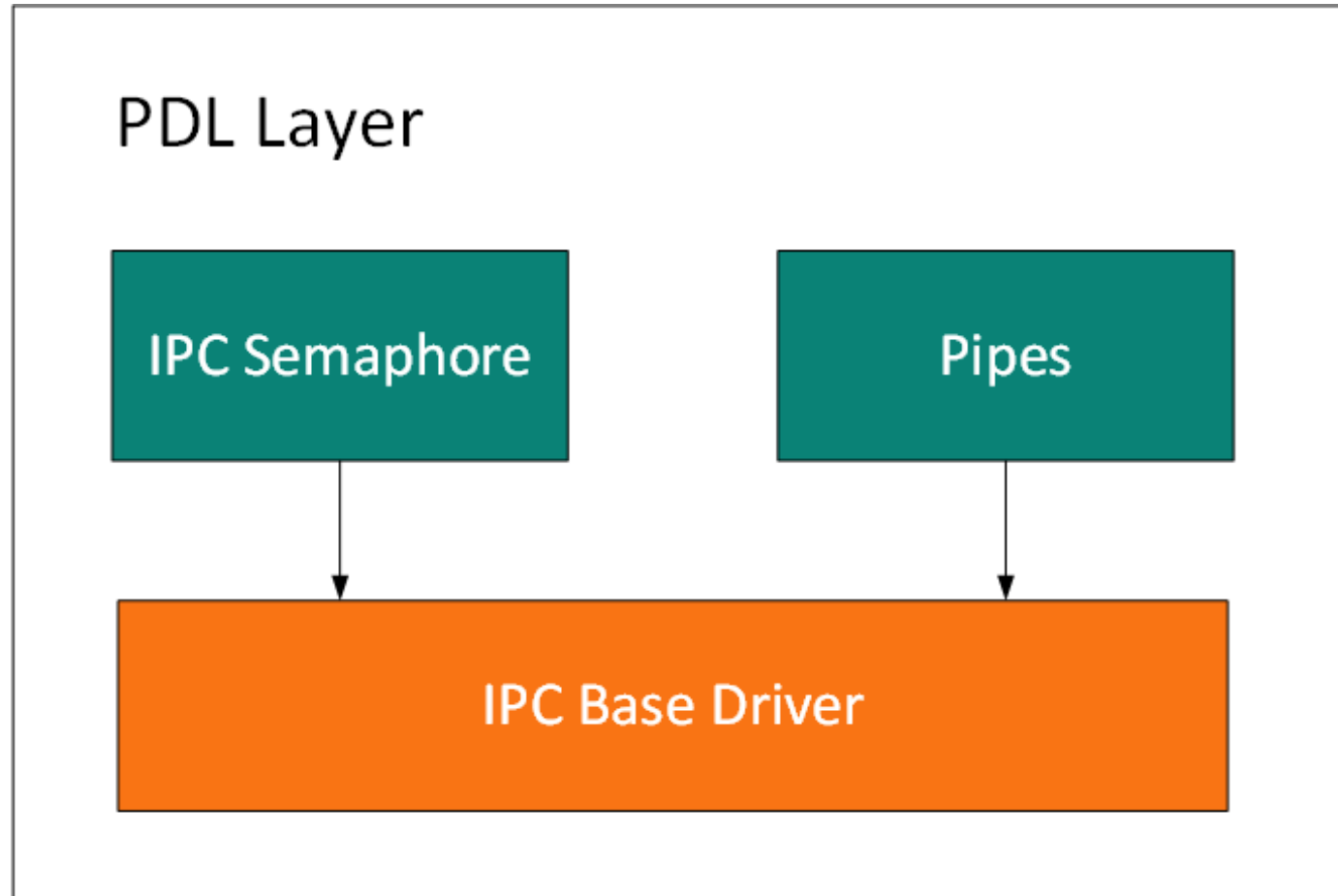
1	Development ecosystem for PSOC™ Edge E84	3
2	Code Example #1: Hello World + Blinky	45
3	Code Example #2: IPC Semaphores using PDL API	49

Overview of IPC

- Supports 16 communication channels
- Supports 16 interrupt structures
- Inter processor communication methods
 - Half-duplex with shared event handling
 - Half-duplex with independent event handling
 - Full-duplex communication
- IPC is available in CPU active power mode only.



Block Diagram of IPC PDL



IPC PDL (Peripheral Abstraction Layer) Functions

Functions

PDL API Reference » IPC (Inter Process Communication) » IPC semaphores layer (IPC_SEMA)

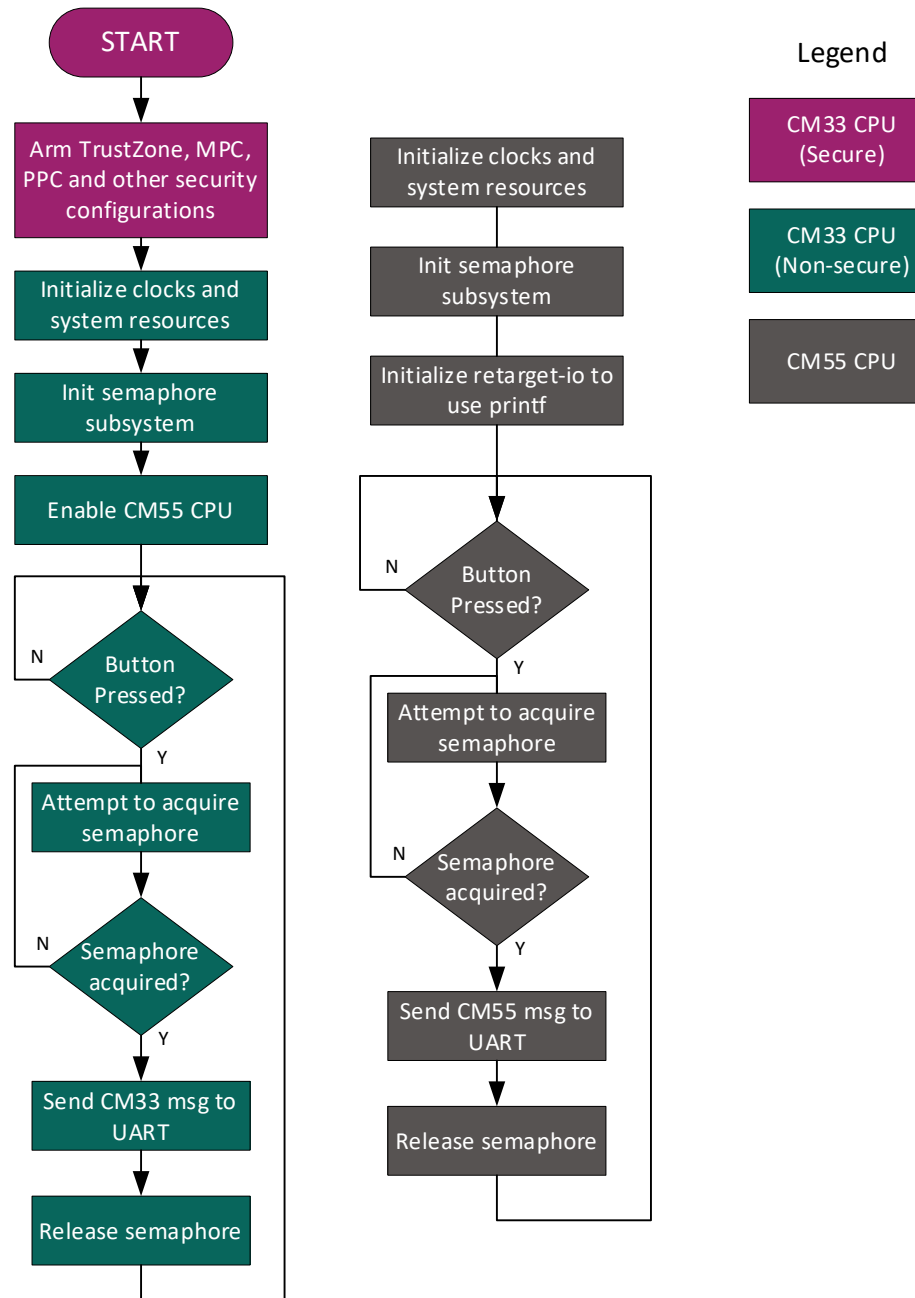
General Description

Functions are used in the driver.

Functions

<code>cy_en_ipcsema_status_t</code>	<code>Cy_IPC_Sema_Init</code> (uint32_t ipcChannel, uint32_t count, uint32_t memPtr[]) This function initializes the semaphores subsystem. More...
<code>cy_en_ipcsema_status_t</code>	<code>Cy_IPC_Sema_InitExt</code> (uint32_t ipcChannel, <code>cy_stc_ipc_sema_t</code> *ipcSema) This function initializes the semaphores subsystem. More...
<code>cy_en_ipcsema_status_t</code>	<code>Cy_IPC_Sema_Set</code> (uint32_t semaNumber, bool preemptable) This function tries to acquire a semaphore. More...
<code>cy_en_ipcsema_status_t</code>	<code>Cy_IPC_Sema_Clear</code> (uint32_t semaNumber, bool preemptable) This functions tries to releases a semaphore. More...
<code>cy_en_ipcsema_status_t</code>	<code>Cy_IPC_Sema_Status</code> (uint32_t semaNumber) This function returns the status of the semaphore. More...
<code>uint32_t</code>	<code>Cy_IPC_Sema_GetMaxSems</code> (void) This function returns the number of semaphores in the semaphores subsystem. More...

Flow chart



Revision history

Document revision	Date	Description of change
**	2025-12-12	Initial release.

